



US 20250285453A1

(19) **United States**

(12) **Patent Application Publication**
Chan et al.

(10) **Pub. No.: US 2025/0285453 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **APPARATUSES, SYSTEMS AND METHODS
FOR GENERATING A VEHICLE DRIVER
SIGNATURE**

continuation of application No. 16/420,345, filed on
May 23, 2019, now Pat. No. 11,281,920.

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Publication Classification

(51) **Int. Cl.**
G06V 20/59 (2022.01)
G06Q 40/08 (2012.01)

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(52) **U.S. Cl.**
CPC **G06V 20/597** (2022.01); **G06Q 40/08**
(2013.01)

(21) Appl. No.: **19/216,204**

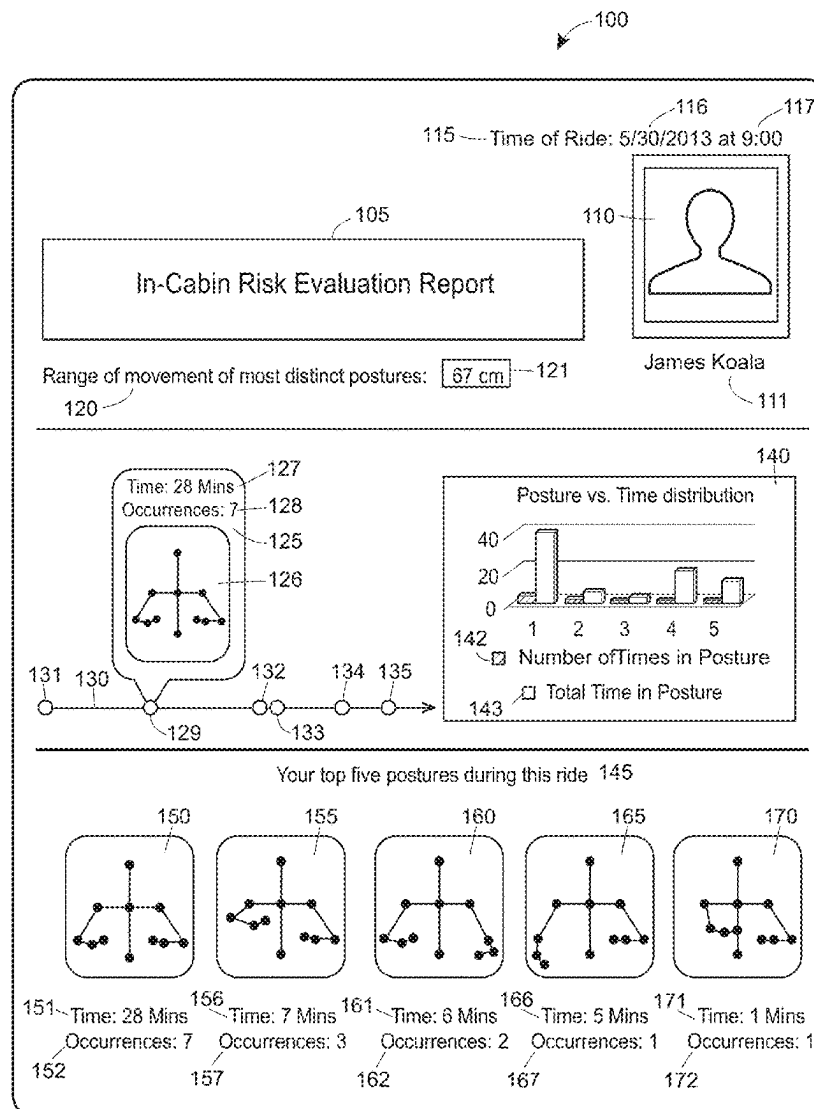
(57) **ABSTRACT**

(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/671,727, filed on
Feb. 15, 2022, now Pat. No. 12,340,602, which is a

Apparatuses, systems and methods are provided for gener-
ating a vehicle driver signature. More particularly, appara-
tuses, systems and methods are provided for generating a
vehicle driver signature based on current image data, pre-
viously classified image data, and vehicle dynamics data.



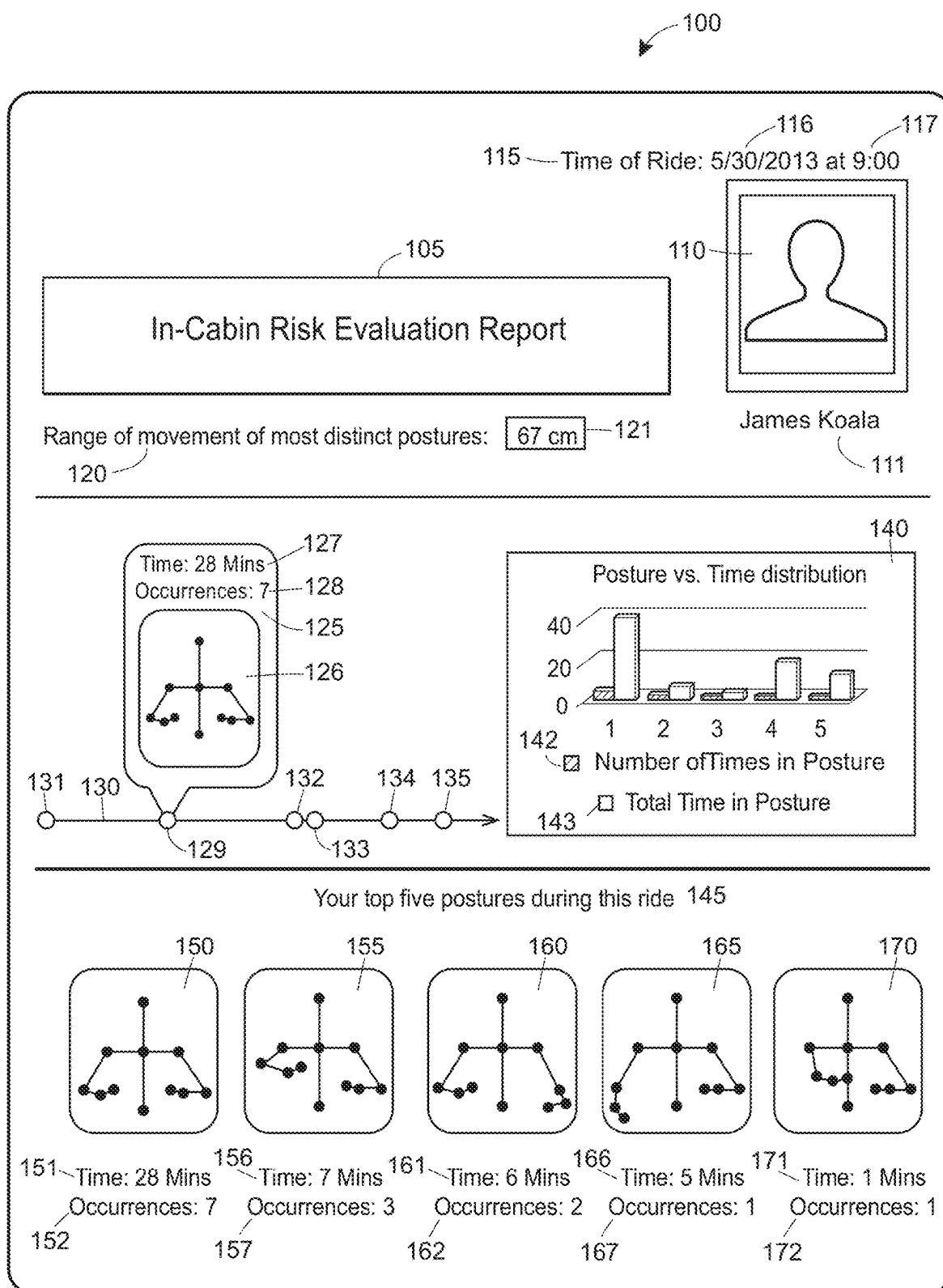


FIG. 1

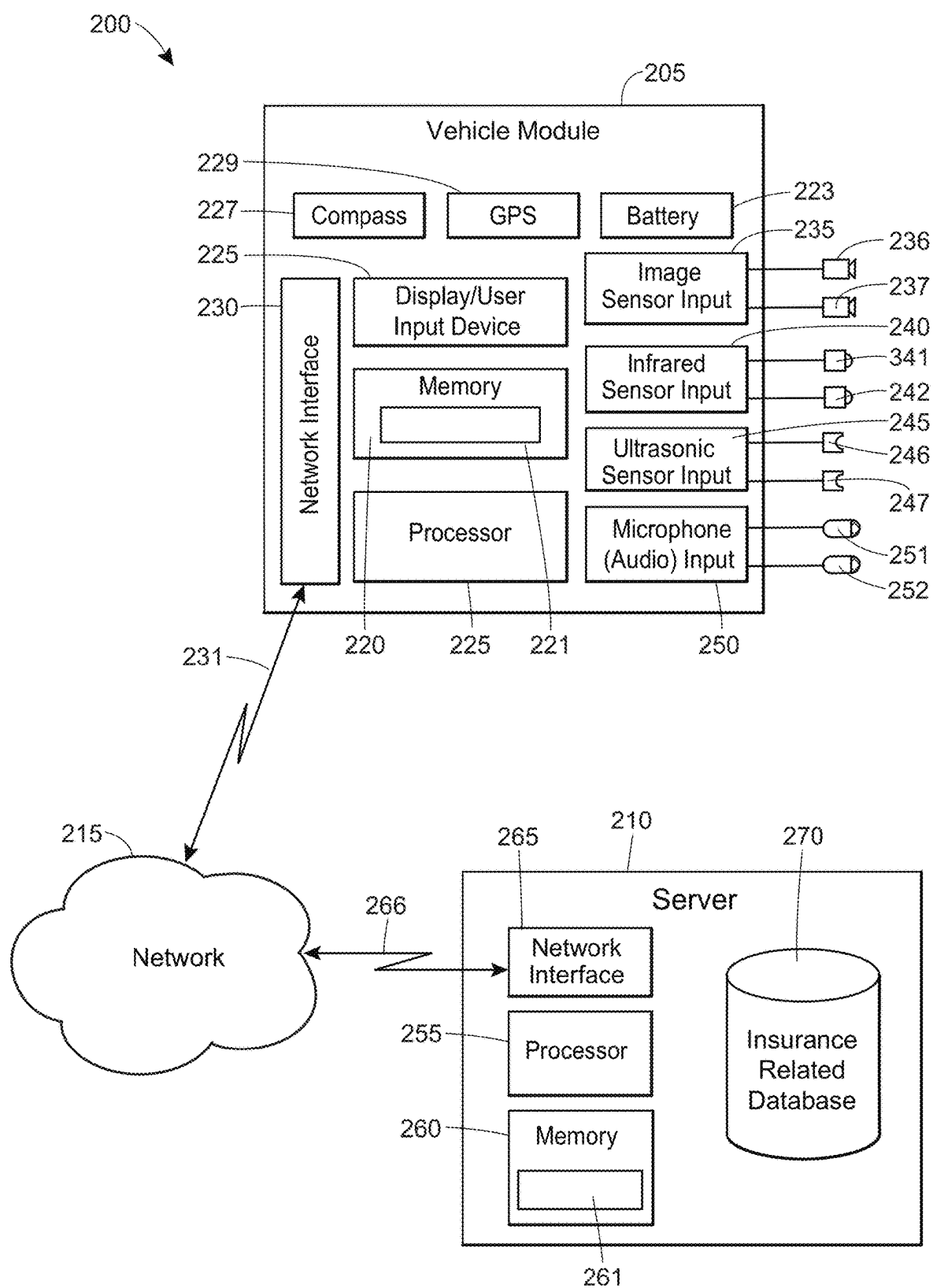


FIG. 2

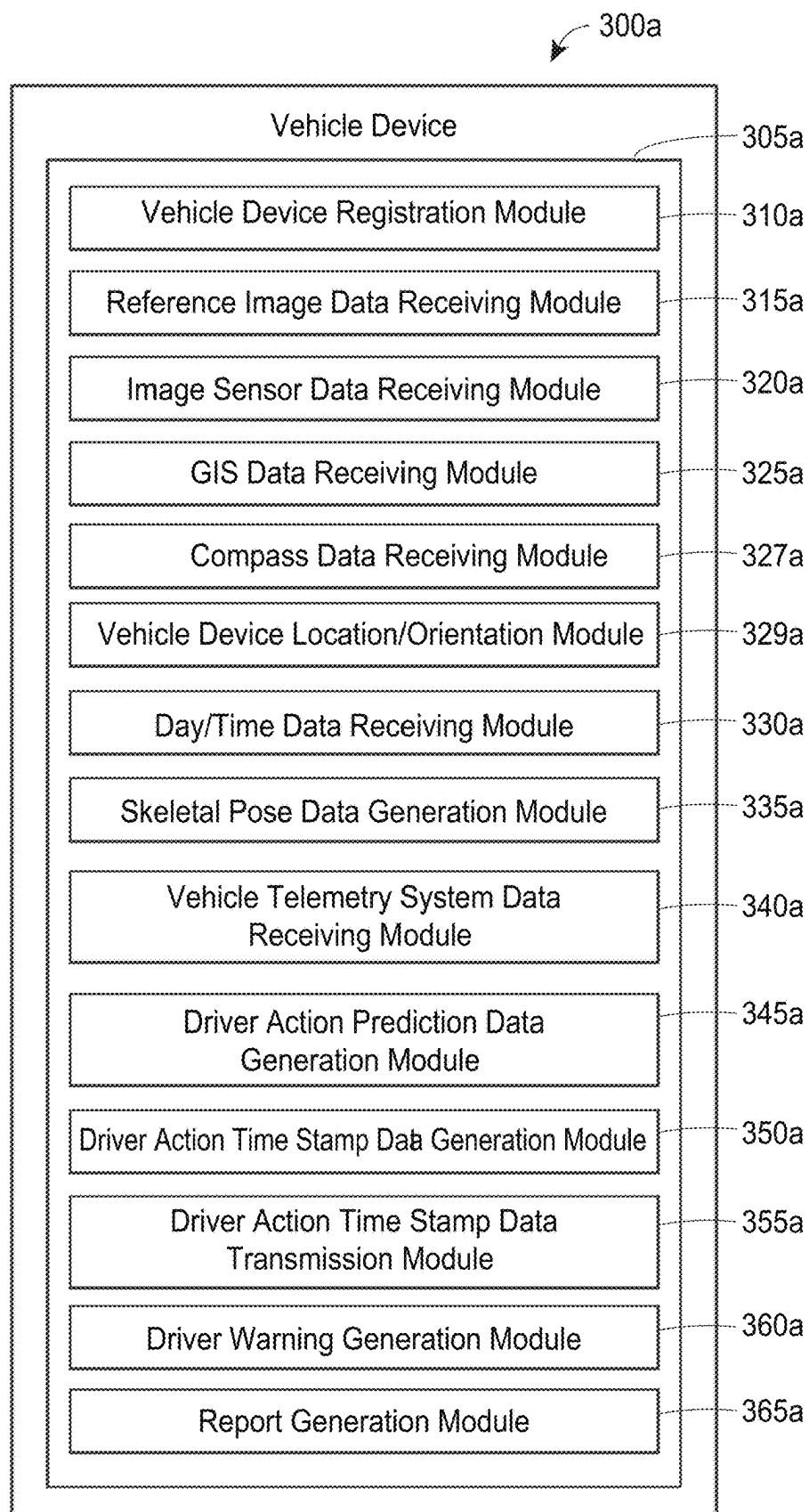
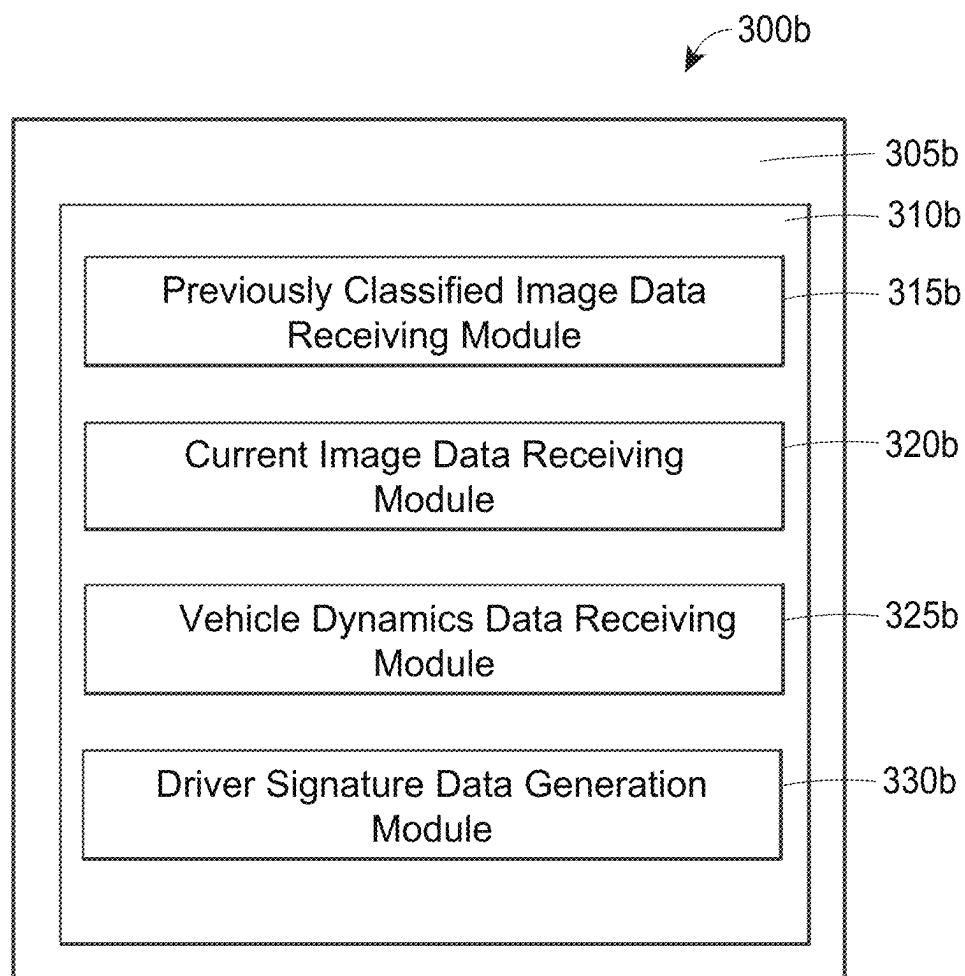
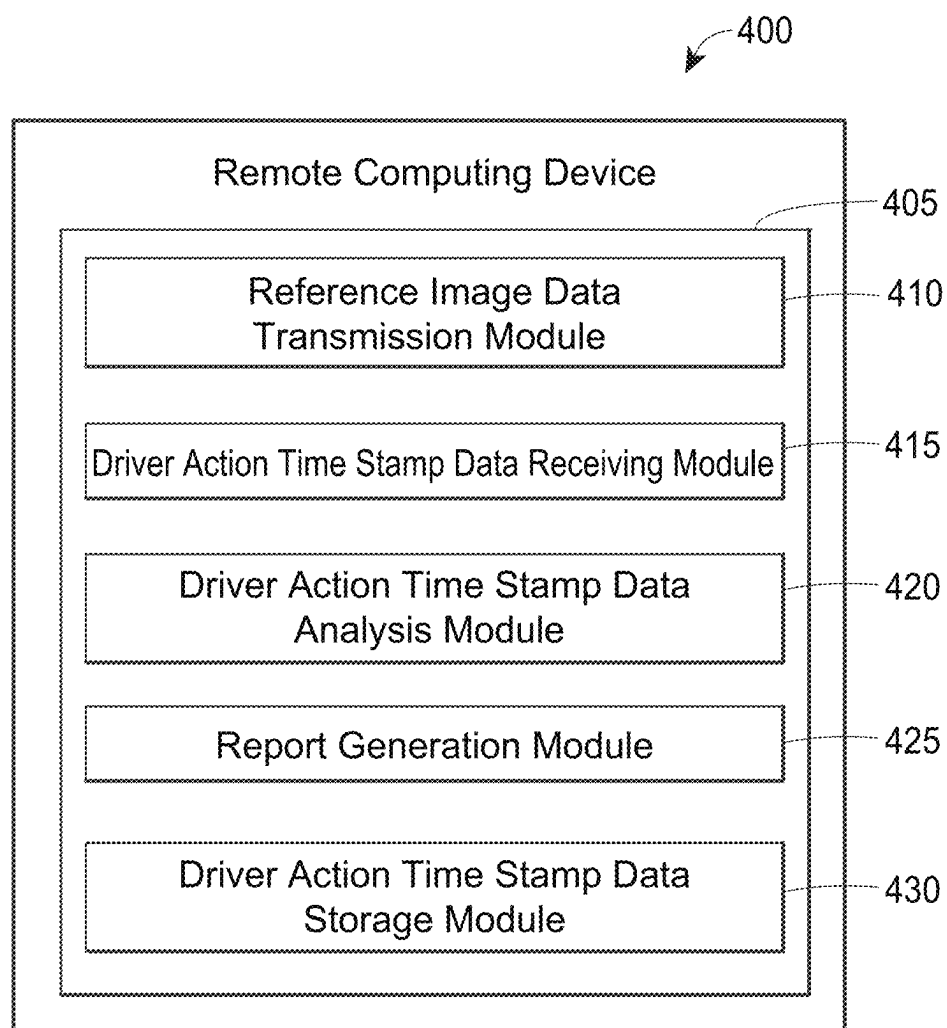
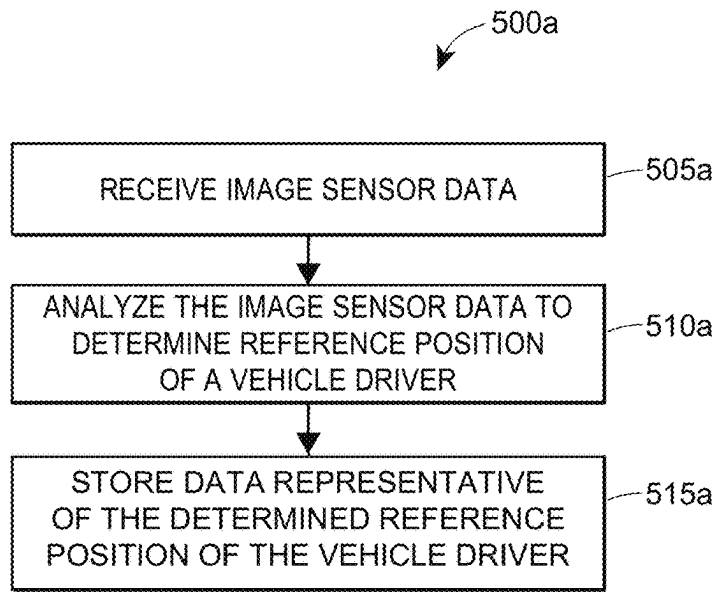
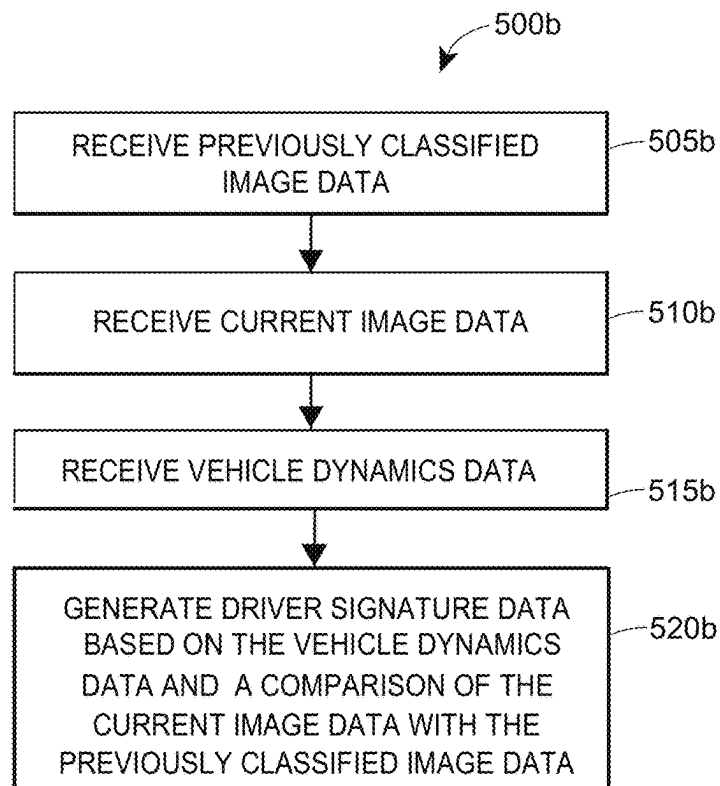


FIG. 3A

**FIG. 3B**

**FIG. 4**

**FIG. 5A****FIG. 5B**

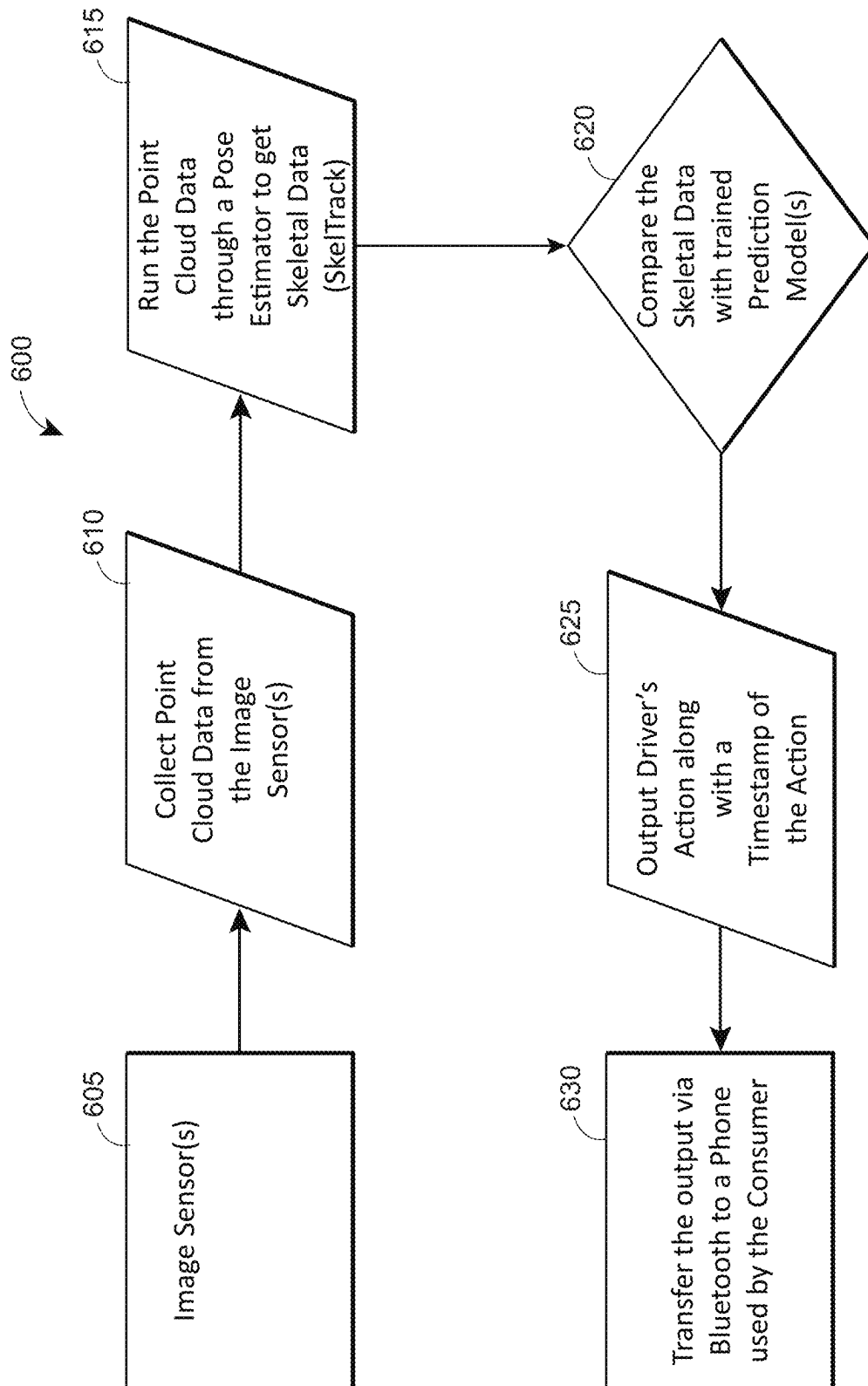


FIG. 6

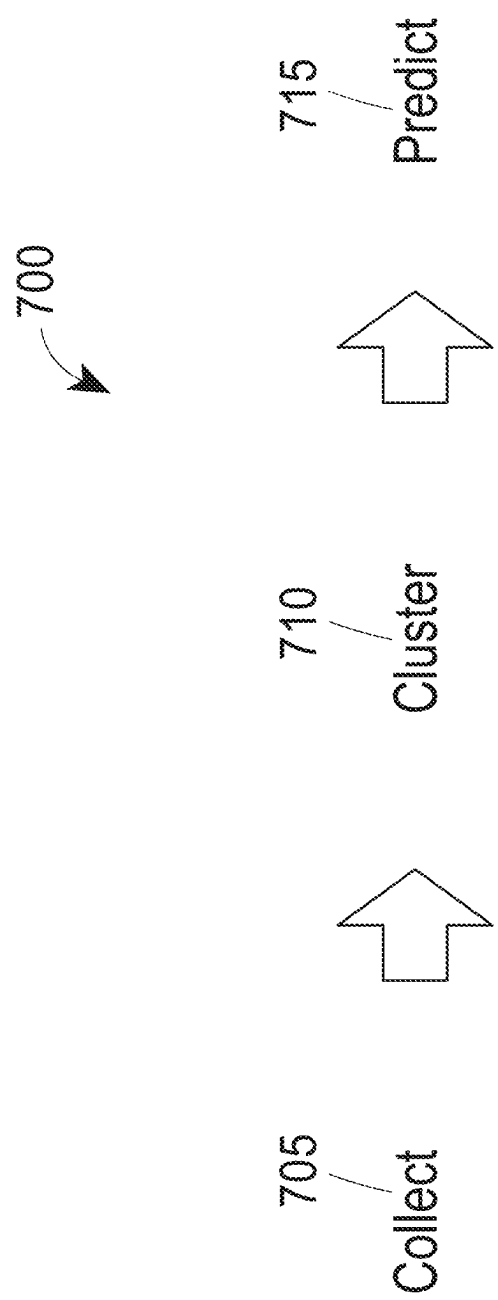


FIG. 7

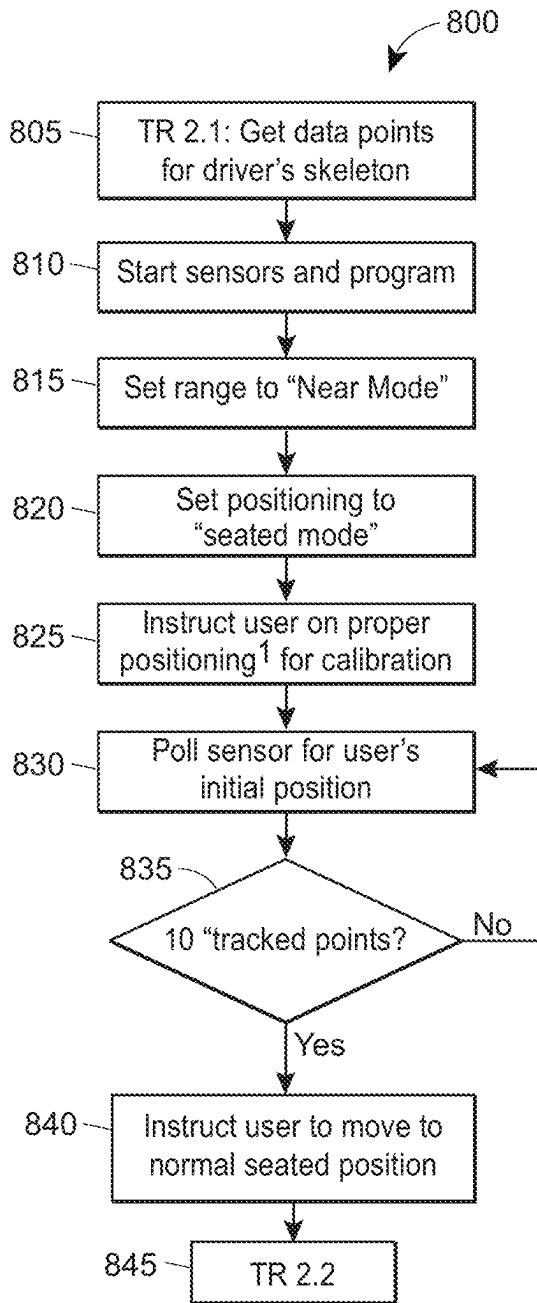


FIG. 8

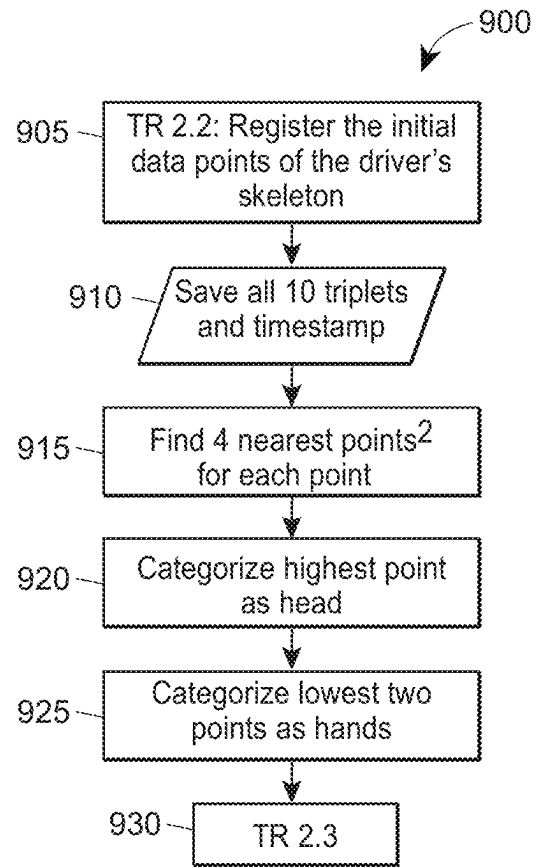
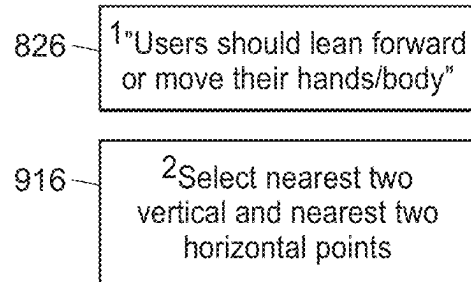


FIG. 9



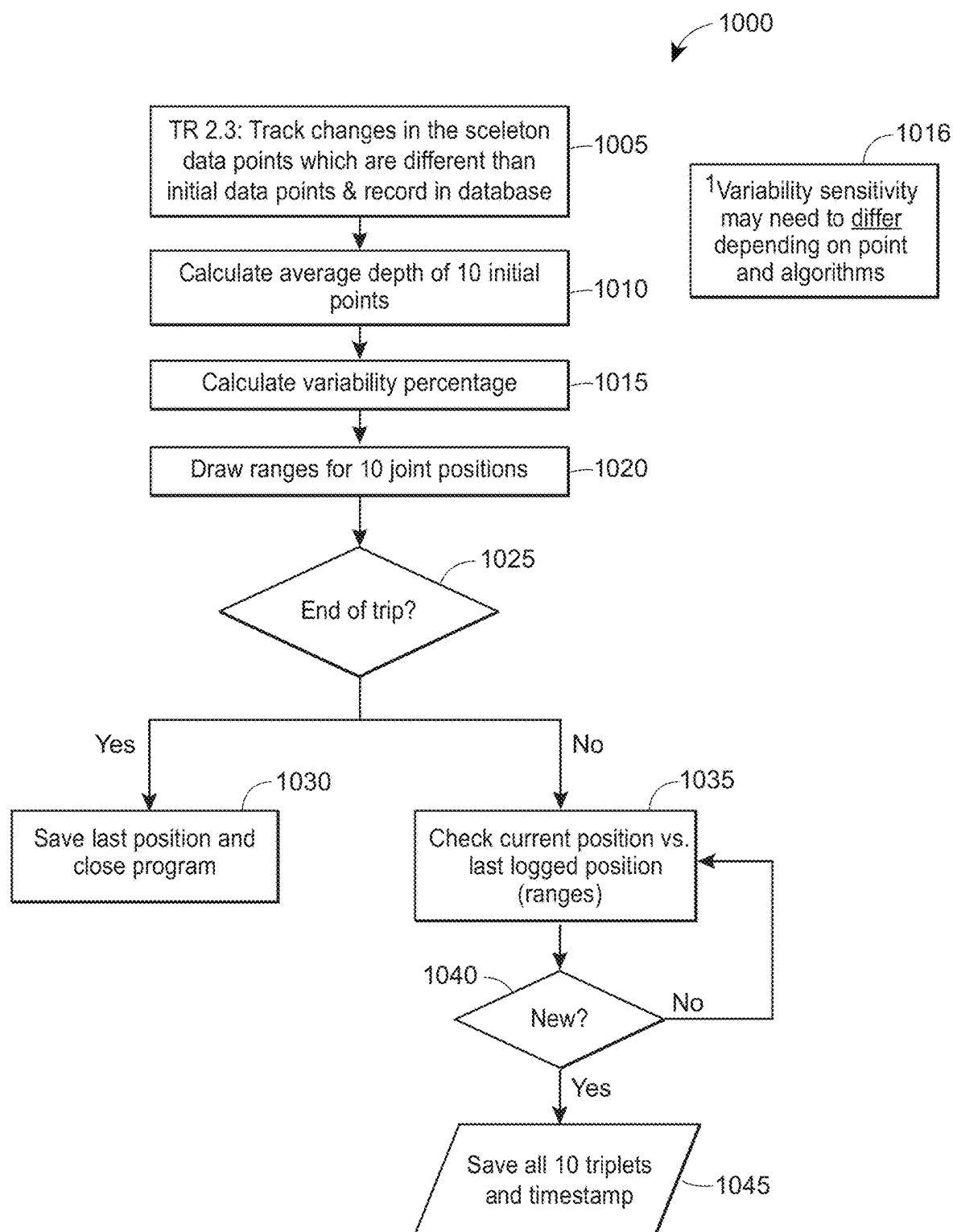


FIG. 10

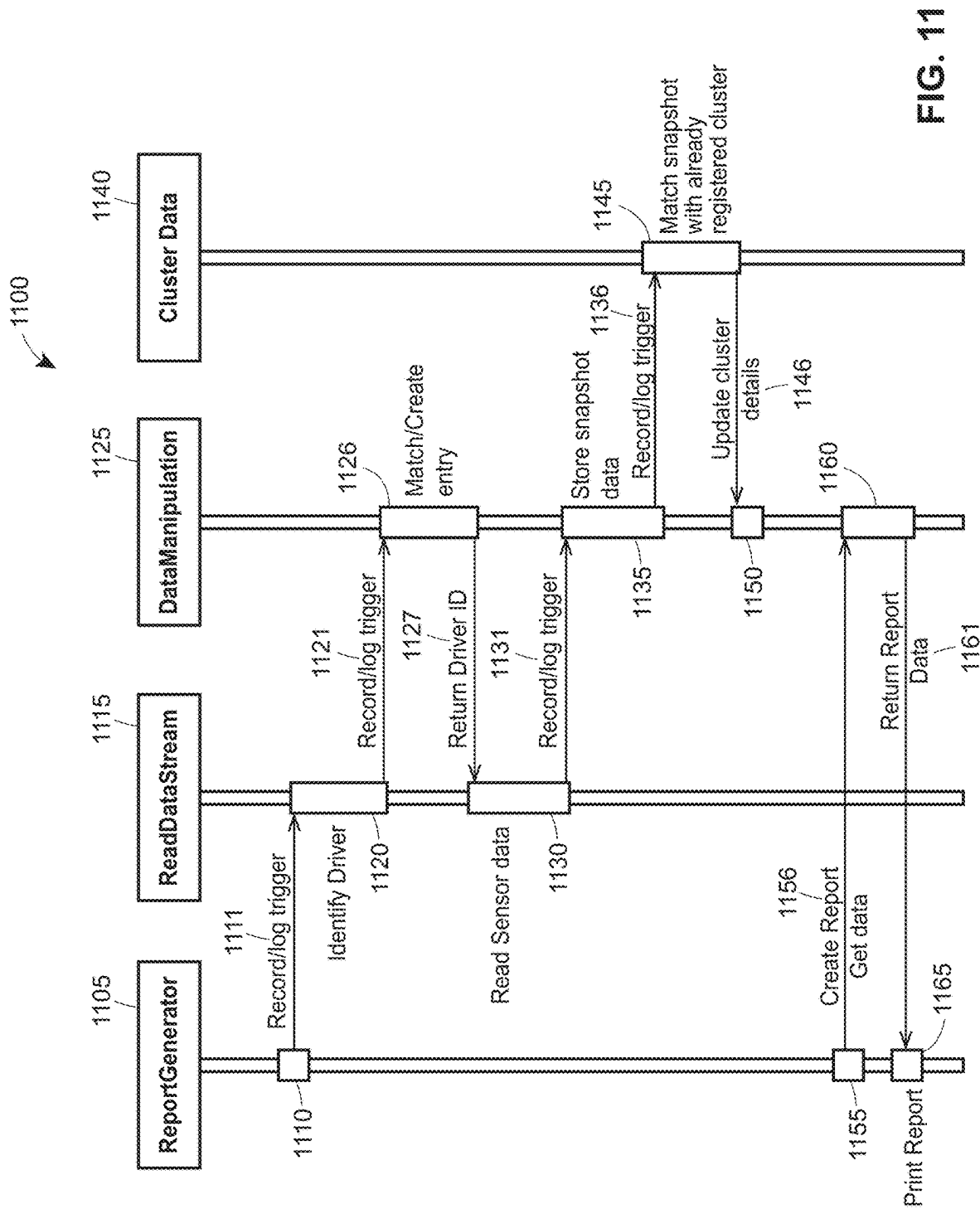


FIG. 11

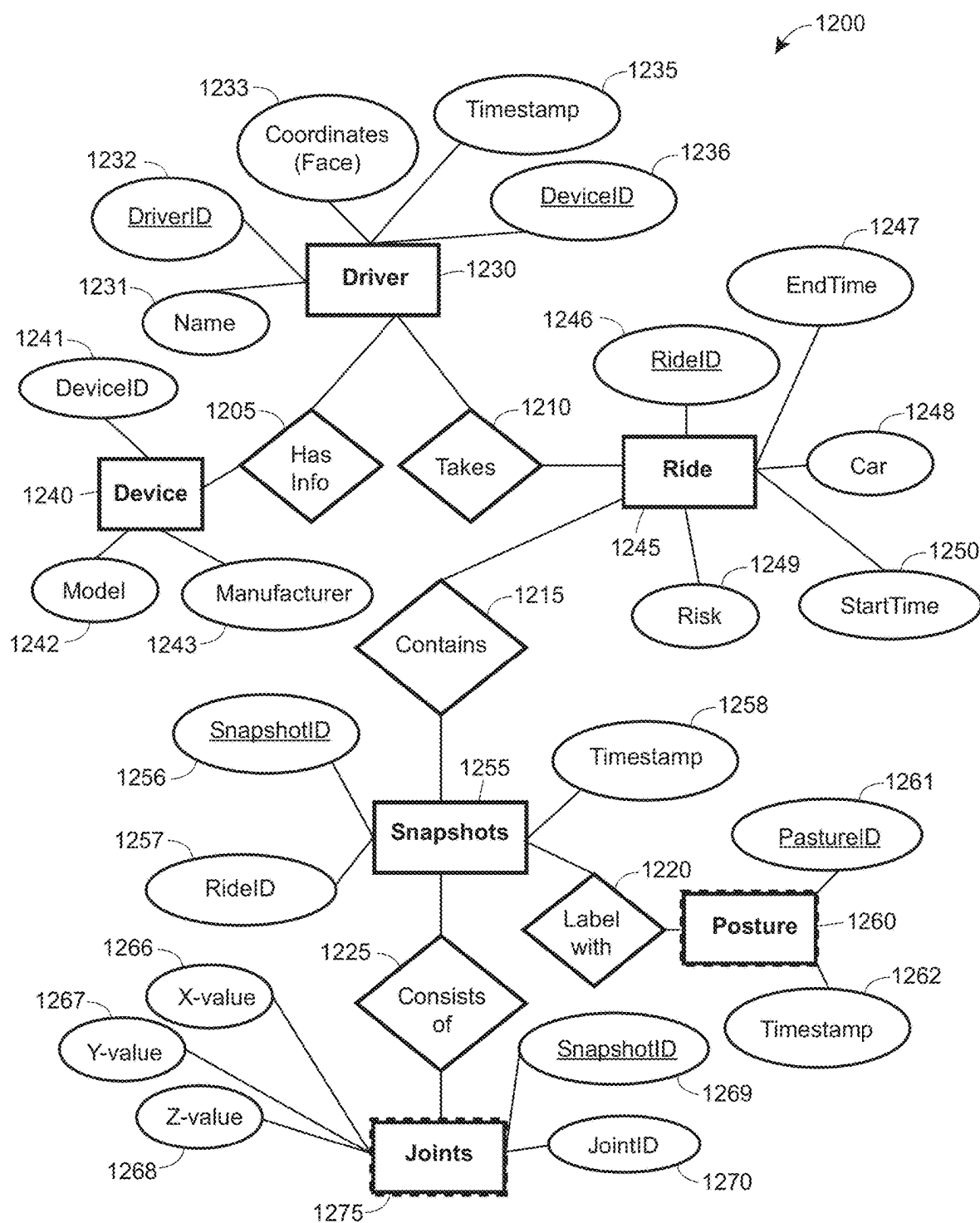


FIG. 12

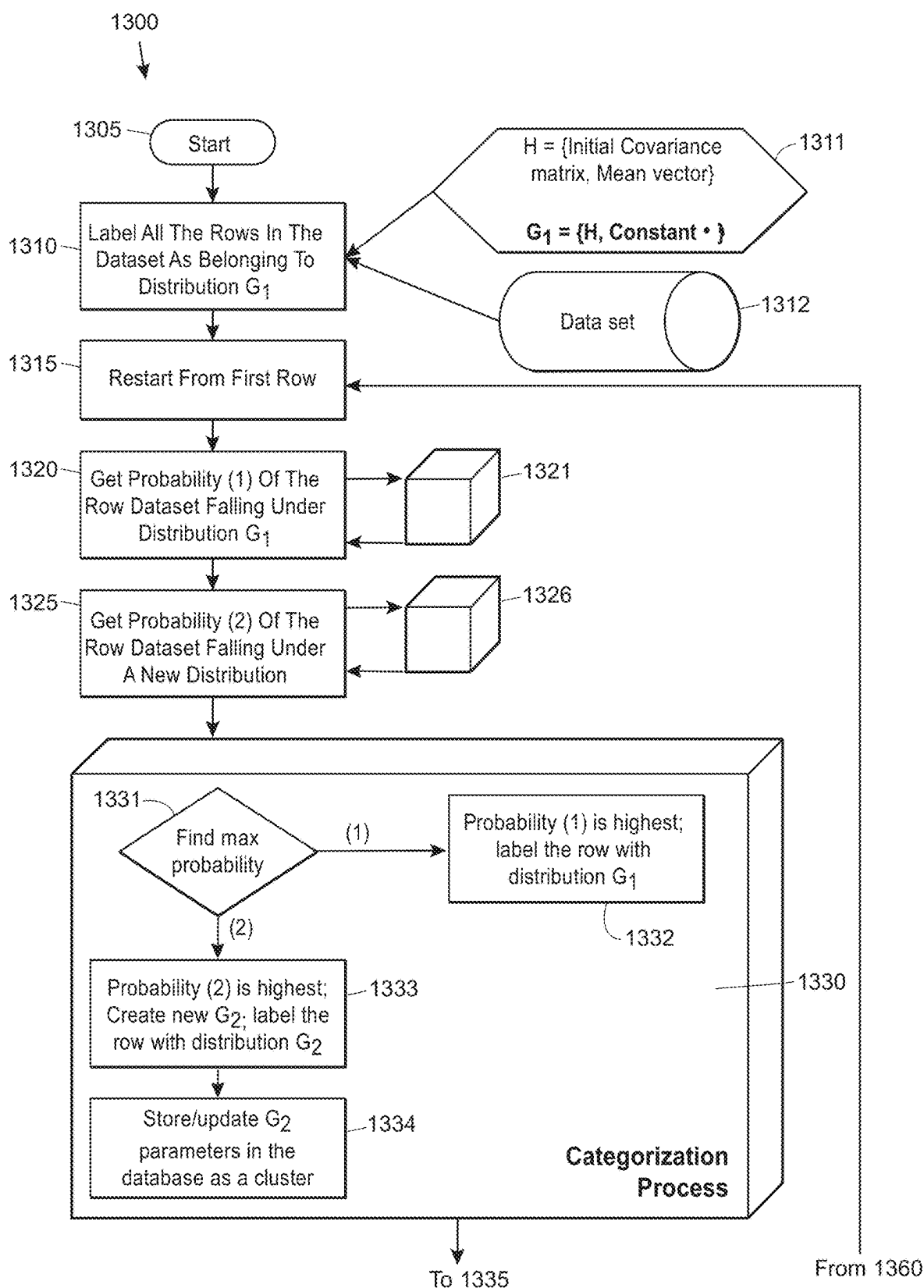


FIG. 13A

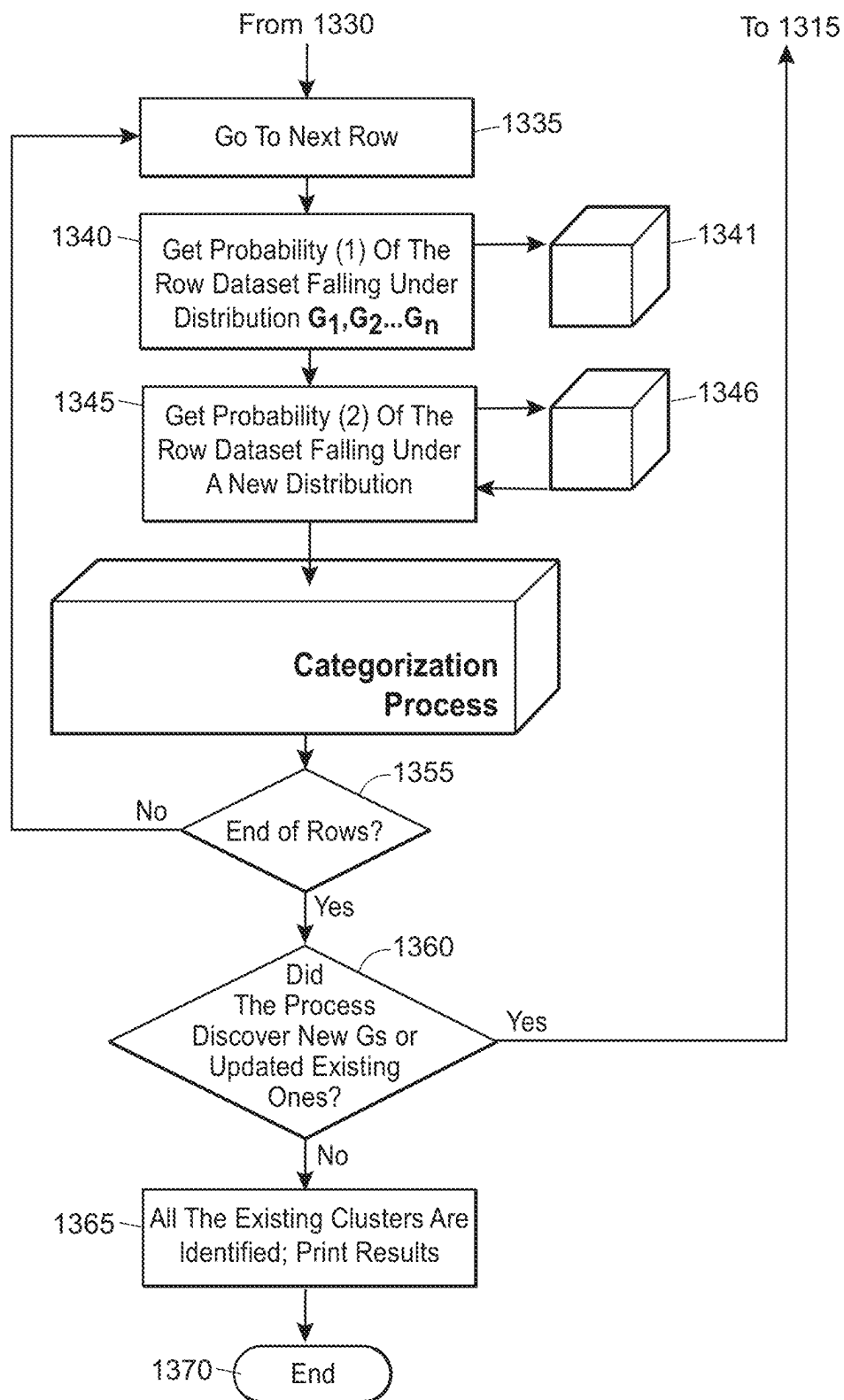


FIG. 13B

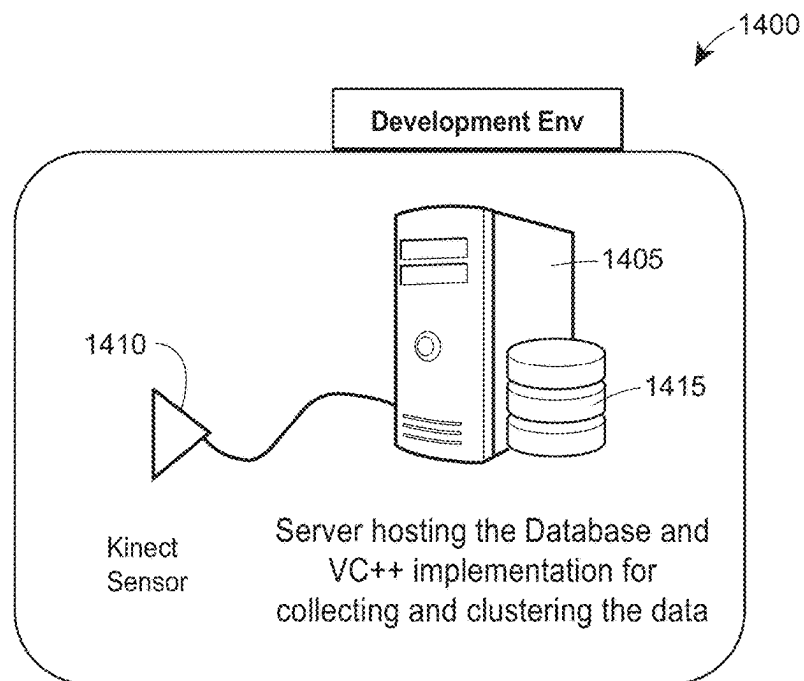


FIG. 14

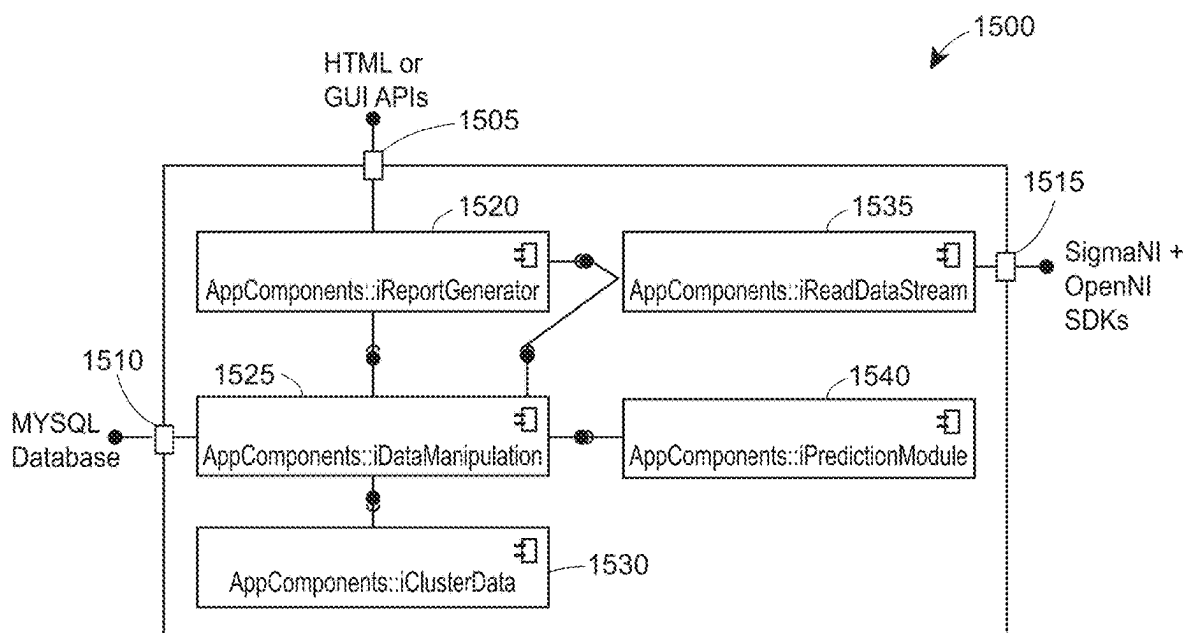


FIG. 15

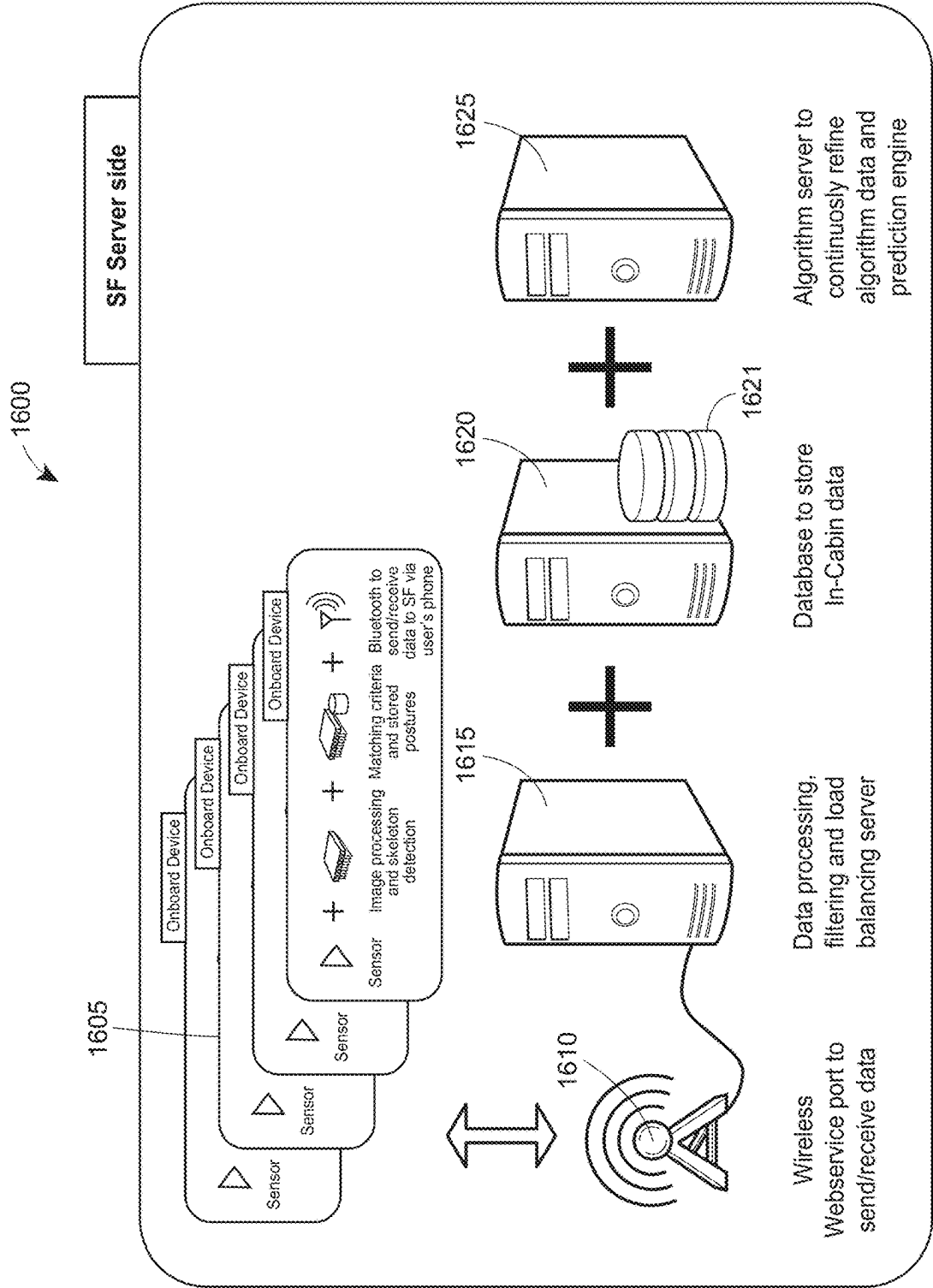


FIG. 16

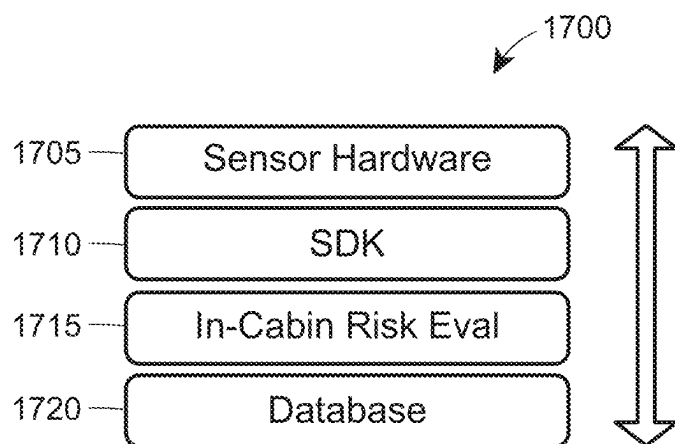


FIG. 17

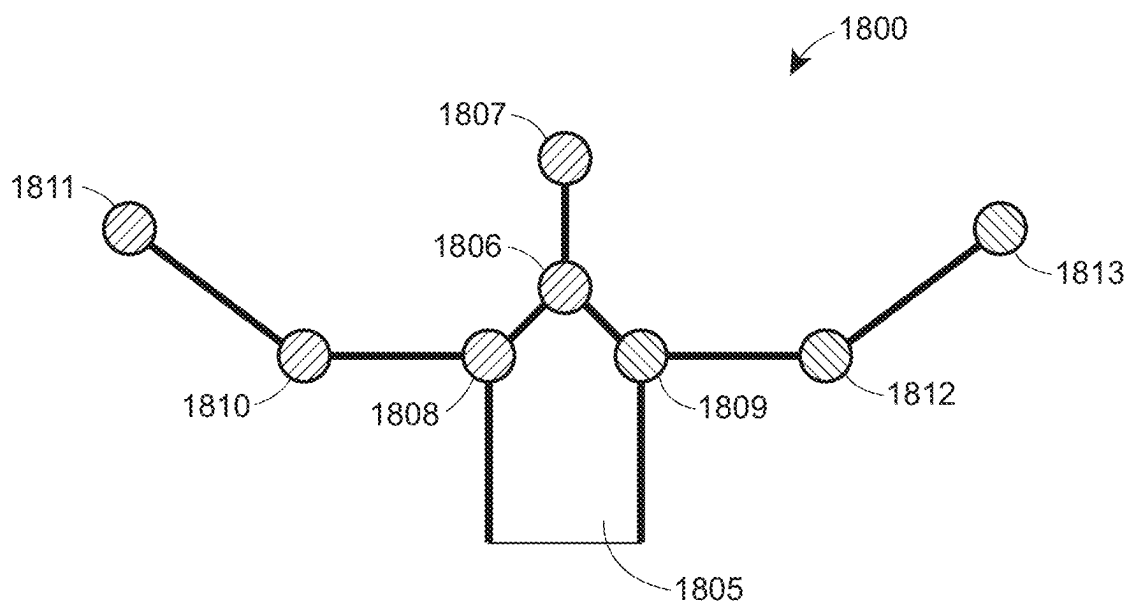


FIG. 18

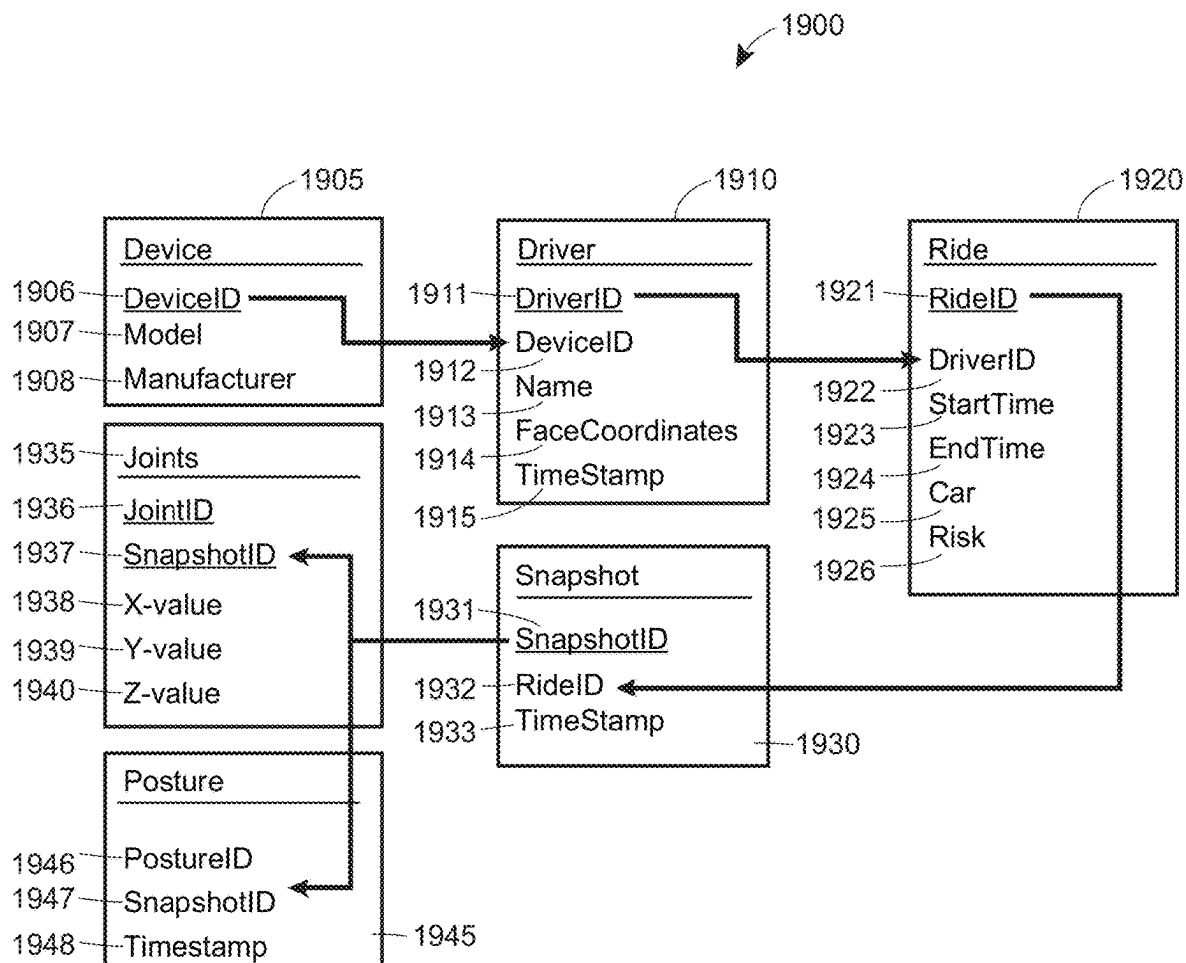


FIG. 19

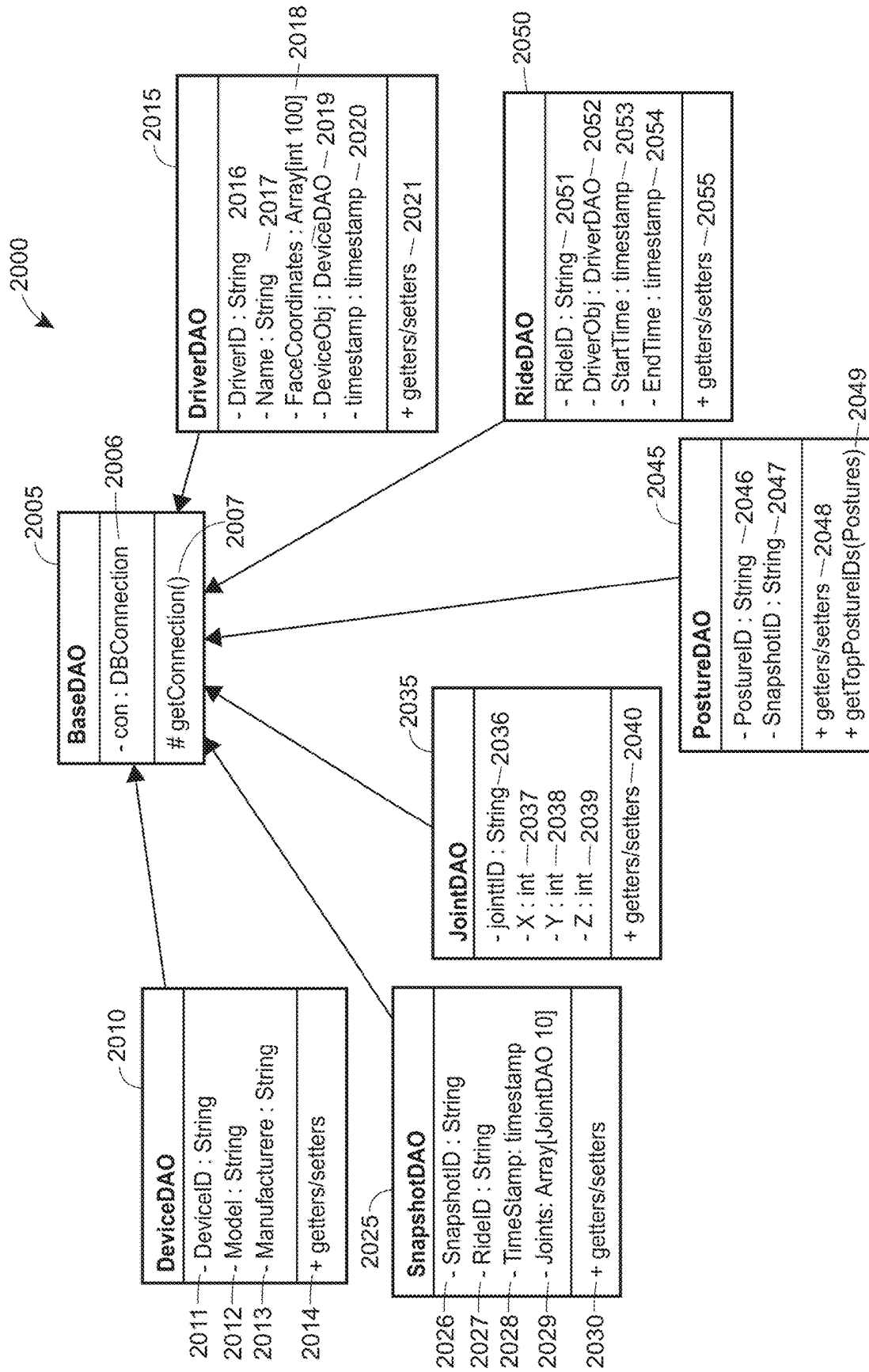


FIG. 20

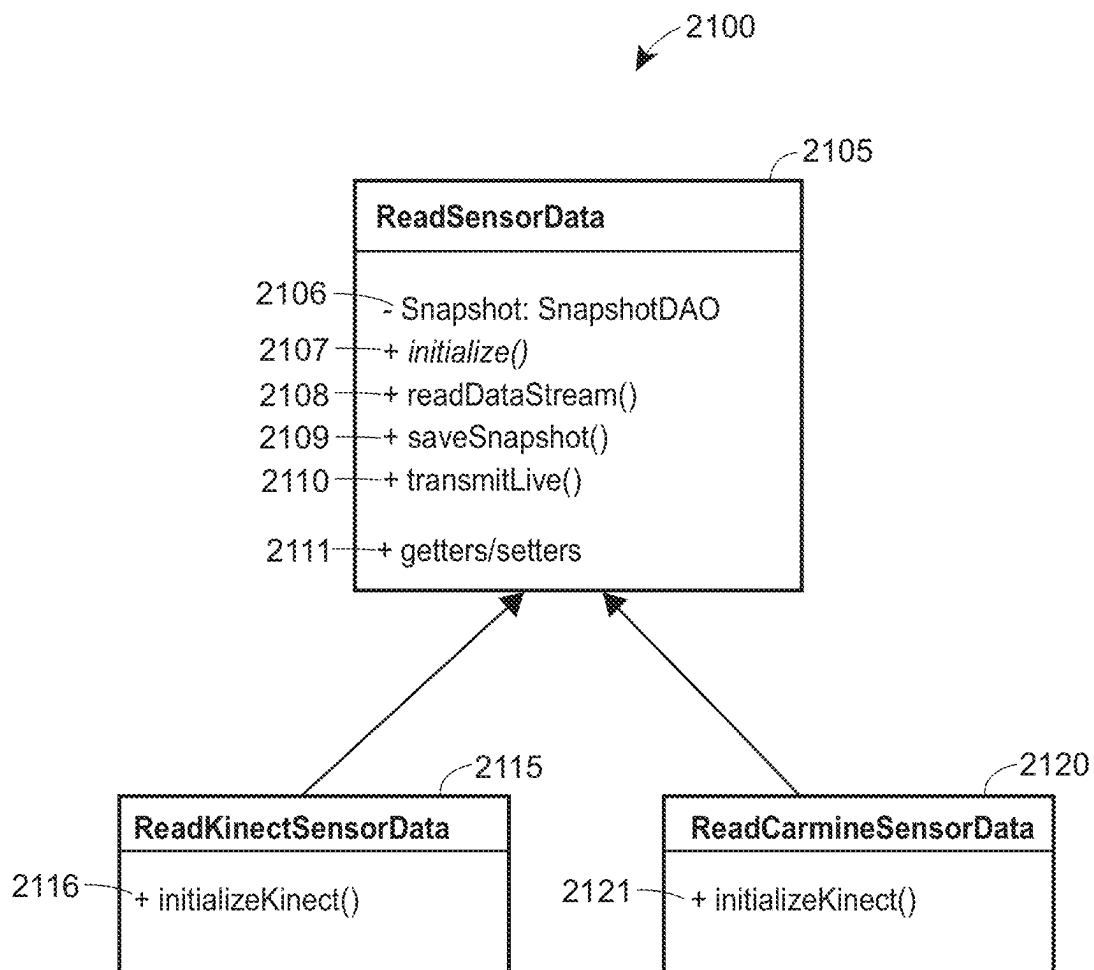


FIG. 21

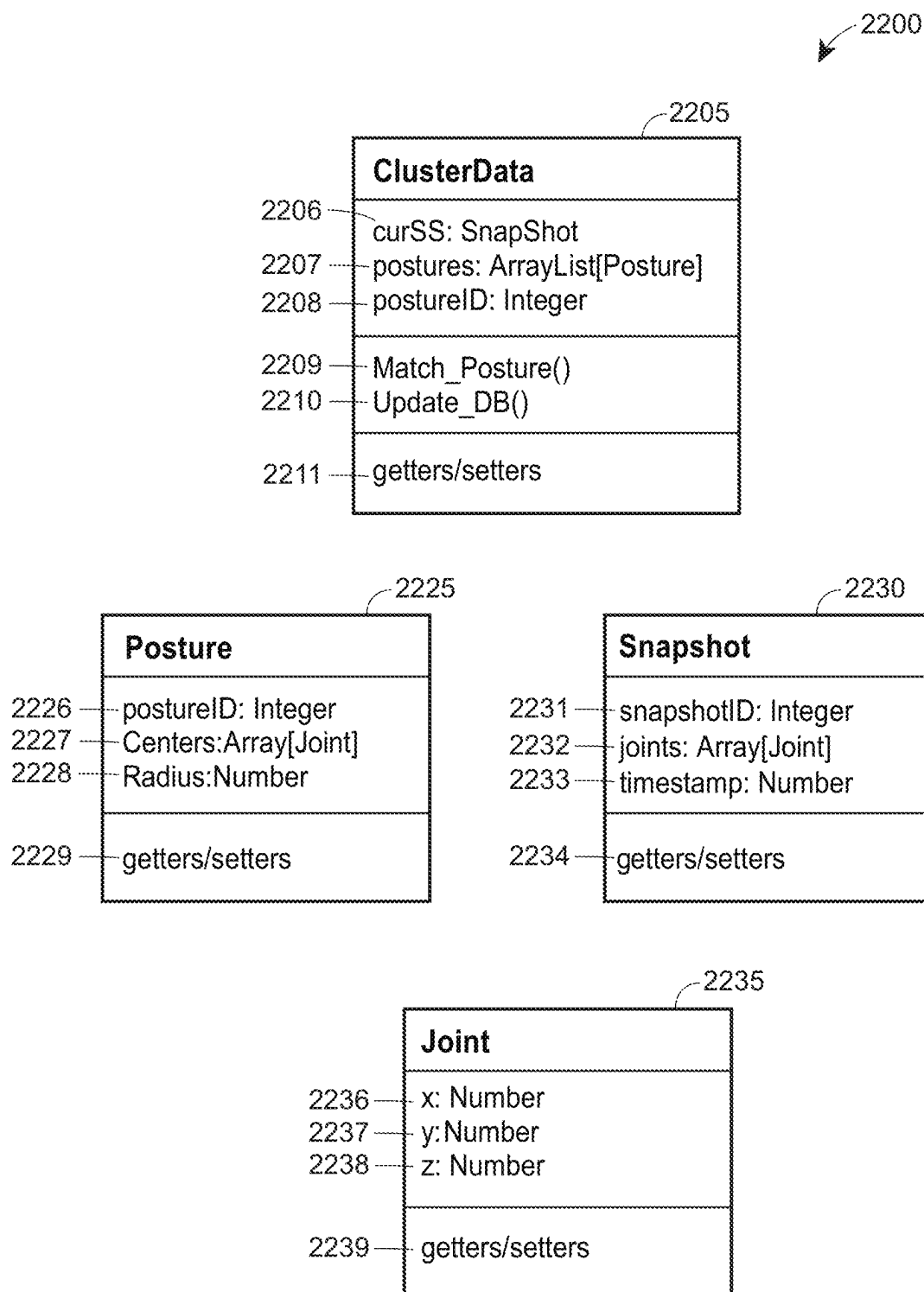


FIG. 22

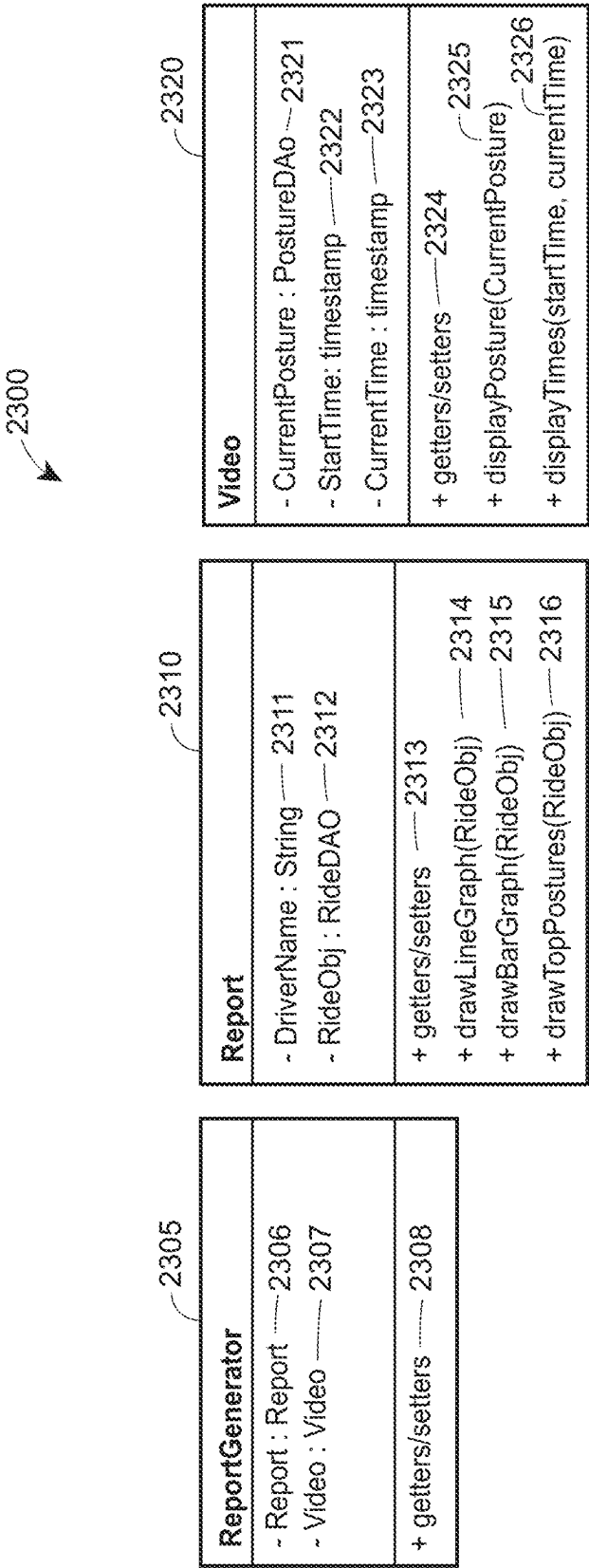


FIG. 23

APPARATUSES, SYSTEMS AND METHODS FOR GENERATING A VEHICLE DRIVER SIGNATURE

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application is a continuation of U.S. patent application Ser. No. 17/671,727 entitled “APPARATUSES, SYSTEMS AND METHODS FOR GENERATING A VEHICLE DRIVER SIGNATURE” filed Feb. 15, 2022, which is a continuation of U.S. patent application Ser. No. 16/420,345 entitled “APPARATUSES, SYSTEMS AND METHODS FOR GENERATING A VEHICLE DRIVER SIGNATURE” filed May 23, 2019, the disclosures of which are hereby incorporated herein by reference in its entirety. The present application is related to U.S. patent application Ser. Nos: 14/994,299, entitled APPARATUSES, SYSTEMS AND METHODS FOR ACQUIRING IMAGES OF OCCUPANTS INSIDE A VEHICLE, filed Jan. 13, 2016; Ser. No. 14/994,305, entitled APPARATUSES, SYSTEMS AND METHODS FOR CLASSIFYING DIGITAL IMAGES, filed Jan. 13, 2016; Ser. No. 14/994,308, entitled APPARATUSES, SYSTEMS AND METHODS FOR CLASSIFYING DIGITAL IMAGES, filed Jan. 13, 2016; Ser. No. 14/994,310, entitled APPARATUSES, SYSTEMS AND METHODS FOR COMPRESSING IMAGE DATA THAT IS REPRESENTATIVE OF A SERIES OF DIGITAL IMAGES, filed Jan. 13, 2016; Ser. No. 14/994,409, entitled APPARATUSES, SYSTEMS AND METHODS FOR DETERMINING DISTRACTIONS ASSOCIATED WITH VEHICLE DRIVING ROUTES, filed Jan. 13, 2016; Ser. No. 14/994,415, entitled APPARATUSES, SYSTEMS AND METHODS FOR GENERATING DATA REPRESENTATIVE OF VEHICLE DRIVER RATINGS, filed Jan. 13, 2016; Ser. No. 14/994,419, entitled APPARATUSES, SYSTEMS AND METHODS FOR GENERATING DATA REPRESENTATIVE OF VEHICLE OCCUPANT POSTURES, filed Jan. 13, 2016; Ser. No. 14/994,424, entitled APPARATUSES, SYSTEMS AND METHODS FOR TRANSITIONING BETWEEN AUTONOMOUS AND MANUAL MODES OF VEHICLE OPERATION, filed Jan. 13, 2016; Ser. No. 14/994,431, entitled APPARATUSES, SYSTEMS AND METHODS FOR DETERMINING WHETHER A VEHICLE IS BEING OPERATED IN AUTONOMOUS MODE OR MANUAL MODE, filed Jan. 13, 2016; Ser. No. 14/994,436, entitled APPARATUSES, SYSTEMS AND METHODS FOR DETERMINING VEHICLE OPERATOR DISTRACTIONS, filed Jan. 13, 2016; Ser. No. 14/994,440, entitled APPARATUSES, SYSTEMS AND METHODS FOR DETERMINING WHETHER A VEHICLE SYSTEM IS DISTRACTING TO A VEHICLE OPERATOR, filed Jan. 13, 2016; 14/862,949, entitled SYSTEMS AND METHODS FOR USING IMAGE DATA TO GENERATE VEHICLE OPERATION LOGS, filed Sep. 23, 2015; and Ser. No. 14/989,524, entitled SYSTEMS AND METHODS FOR ASSOCIATING VEHICLE OPERATORS WITH DRIVING MISSES INDICATED IN VEHICLE OPERATION DATA, filed Jan. 6, 2016; the disclosures of which are incorporated herein in their entireties by reference thereto.

TECHNICAL FIELD

[0002] The present disclosure is directed to apparatuses, systems and methods for generating a vehicle driver signature.

More particularly, the present disclosure is directed to apparatuses, systems and methods for generating a vehicle driver signature based on vehicle interior image data and vehicle dynamics.

BACKGROUND

[0003] Vehicles are being provided with more complex systems. For example, vehicles commonly include a plethora of entertainment systems, such as stereos, USB interfaces for mobile telephones, video players, etc. Vehicles often have a host of other operator interfaces, such as emergency calling systems, vehicle navigation systems, heating and air conditioning systems, interior and exterior lighting controls, air bags, seatbelts, etc.

[0004] Vehicle operating environments are becoming more complex as well. For example, some roadways include U-turn lanes, round-a-bouts, no-left turn, multiple lanes one way in the morning and the other way in the afternoon, etc. Increases in traffic are also contributing to increased complexity.

[0005] These additional complexities contribute to increases in driver distractions. What is needed are methods and systems for generating a vehicle driver signature based on vehicle interior image data.

SUMMARY

[0006] A device for determining a driver signature of a driver includes a processor and a memory including instructions that, when executed by the processor, cause the processor to: receive current image data from at least one sensor, wherein the current image data is representative of one or more postures of the driver currently within a vehicle; receive previously classified image data, wherein the previously classified image data is representative of known postures associated with the driver; receive vehicle dynamics data, wherein the vehicle dynamics data is representative of a driving behavior of the driver; and generate the driver signature that identifies the driver based on the vehicle dynamics data and a comparison of the current image data with the previously classified image data.

[0007] A computer-implemented method for determining a driver signature of a driver includes: receiving, at a processor, current image data from at least one sensor, wherein the current image data is representative of one or more postures of the driver currently within a vehicle; receiving, at the processor, previously classified image data, wherein the previously classified image data is representative of known postures associated with the driver; receiving, at the processor, vehicle dynamics data, wherein the vehicle dynamics data is representative of a driving behavior of the driver; and generating, using the processor, the driver signature that identifies the driver based on the vehicle dynamics data and a comparison of the current image data with the previously classified image data.

[0008] A non-transitory computer-readable medium storing computer-readable instructions that, when executed by a processor, cause the processor to: receive current image data from at least one sensor, wherein the current image data is representative of one or more postures of a driver currently within a vehicle; receive previously classified image data, wherein the previously classified image data is representative of known postures associated with the driver; receive vehicle dynamics data, wherein the vehicle dynamics data is

representative of a driving behavior of the driver; and generate a driver signature that identifies the driver based on the vehicle dynamics data and a comparison of the current image data with the previously classified image data.

[0009] In one embodiment, a device for determining a driver signature may include a vehicle interior data receiving module stored on a memory that, when executed by a processor, causes the processor to receive vehicle interior data from at least one vehicle interior sensor. The vehicle interior data may be representative of driver signatures. The device may also include a previously classified vehicle interior data receiving module stored on the memory that, when executed by the processor, causes the processor to receive previously classified vehicle interior data. The previously classified vehicle interior data may be representative of known driver signatures. The device may further include a vehicle dynamics data receiving module stored on the memory that, when executed by the processor, may cause the processor to receive vehicle dynamics data. The vehicle dynamics data may be representative of operation of a vehicle. The device may yet further include a vehicle driver signature determination module stored on the memory that, when executed by the processor, causes the processor to determine at least one driver signature based on the vehicle dynamics data and a comparison of the vehicle interior data with the previously classified vehicle interior data.

[0010] In another embodiment, a computer-implemented method for determining a driver signature may include receiving, at a processor, vehicle interior data from at least one vehicle interior sensor. The vehicle interior data may be representative of vehicle occupant postures. The method may also include receiving, at the processor, previously classified vehicle interior data. The previously classified vehicle interior data may be representative of known vehicle occupant postures. The method may further include receiving, at the processor, vehicle dynamics data. The vehicle dynamics data may be representative of operation of a vehicle. The method may yet further include determining, using the processor, at least one vehicle driver signature based on the vehicle dynamics data and a comparison of the vehicle interior data with the previously classified vehicle interior data.

[0011] In a further embodiment, a non-transitory computer-readable medium storing computer-readable instructions that, when executed by a processor, cause the processor to determine a driver signature may include a vehicle interior data receiving module that, when executed by a processor, causes the processor to receive vehicle interior data from at least one vehicle interior sensor. The vehicle interior data may be representative of driver signatures. The non-transitory computer-readable medium may also include a previously classified vehicle interior data receiving module that, when executed by the processor, causes the processor to receive previously classified vehicle interior data. The previously classified vehicle interior data may be representative of known driver signatures. The non-transitory computer-readable medium may further include a vehicle dynamics data receiving module, when executed by the processor, may cause the processor to receive vehicle dynamics data. The vehicle dynamics data may be representative of operation of a vehicle. The non-transitory computer-readable medium may yet further include a vehicle driver signature determination module stored on the memory that, when executed by the processor, causes the processor

to determine at least one driver signature based on the vehicle dynamics data and a comparison of the vehicle interior data with the previously classified vehicle interior data.

BRIEF DESCRIPTION OF THE FIGURES

[0012] FIG. 1 depicts an example report for a vehicle in-cabin insurance risk evaluation;

[0013] FIG. 2 depicts a high-level block diagram for an example computer system for generating data representative of vehicle in-cabin insurance risk evaluations;

[0014] FIGS. 3A and 3B depict block diagrams of example vehicle devices for use in generating a vehicle driver signature;

[0015] FIG. 4 depicts a block diagram of an example remote computing device for use in generating data representative of vehicle in-cabin insurance risk evaluations;

[0016] FIGS. 5A and 5B depict flow diagrams for example methods of generating a vehicle driver signature;

[0017] FIG. 6 depicts a flow diagram for an example method of generating data representative of a vehicle driver's actions along with an associated time stamp;

[0018] FIG. 7 depicts a flow diagram for an example method of generating data representative of a prediction of a vehicle driver's action;

[0019] FIGS. 8-10 depict flow diagrams for example methods for tracking movement of a vehicle driver's upper body;

[0020] FIG. 11 depicts an example sequence diagram for generating a report for a vehicle in-cabin insurance risk evaluation;

[0021] FIG. 12 depicts a detailed example E-R diagram for generating data representative of a vehicle driver's actions along with an associated time stamp;

[0022] FIGS. 13A and 13B depict a flow diagram for an example method of a development environment for generating data representative of vehicle in-cabin insurance risk evaluations;

[0023] FIG. 14 depicts an example computer system for development of a model for generating data representative of vehicle in-cabin insurance risk evaluations;

[0024] FIG. 15 depicts a block diagram of various components for development of a model for generating data representative of vehicle in-cabin insurance risk evaluations;

[0025] FIG. 16 depicts a block diagram for an example server side system for generating data representative of vehicle in-cabin insurance risk evaluations;

[0026] FIG. 17 depicts a flow diagram for example method of directly interacting with the SDK to obtain driver data for use in generating data representative of vehicle in-cabin insurance evaluations;

[0027] FIG. 18 depicts an example skeletal figure representative of a vehicle driver's upper body position;

[0028] FIG. 19 depicts an example object design for a database for use in generating data representative of vehicle in-cabin insurance risk evaluations; and

[0029] FIGS. 20-23 depict various example class diagrams of objects for use in a database of FIG. 19.

DETAIL DESCRIPTION

[0030] Apparatuses, systems and methods for generating data representative of vehicle occupant distractions may include the following capabilities: 1) determine whether a

vehicle driver is looking at a road (i.e., tracking the driver's face/eyes, with emphasis on differentiating between similar actions, such as a driver who is adjusting a radio while looking at the road versus adjusting the radio while not looking at the road at all); 2) determine whether a driver's hands are empty (e.g., including determining an approximate size/shape of object in a driver's hands to, for example, differentiate between a cell phone and a large cup, for example); 3) identify a finite number of driver postures; and 4) logging rotated and scaled postures that are normalized for a range of different drivers.

[0031] An associated mobile application may accommodate all popular platforms, such as iOS, Android and Windows, to connect an onboard device to a cell phone. In addition, to act as data connection provider to remote servers, the mobile application may provide a user friendly interface for reporting and troubleshooting. Accordingly, associated memory, processing, and related data transmission requirements are reduced compared to previous approaches.

[0032] The apparatuses, systems and methods of the present disclosure may confirm an identity of a current driver of a given vehicle by conferring between at least two data streams (e.g., previously classified image data and current image data). The image data may be, for example, either 3D point cloud data or 2D image or video data. A "driver signature" may be determined based on a pattern of vehicle dynamics and driving style. A "driver signature" may represent, for example, a "driver ID" and/or a "driver posture" and/or a "driver wireframe" and/or a "driving style" and/or a "driver classification." Alternatively, or additionally, a "driver signature" may represent, for example, a driver ID (or a set of driver characteristics) based on one or more of on vehicle interior image data and vehicle dynamics data. Alternatively, or additionally, a driver signature may be determined based on vehicle interior image data and vehicle dynamics data. The vehicle interior image data may be representative of an identification of a particular individual. The vehicle dynamics data may be representative of a particular driving style, or driving behavior, associated with a particular individual (e.g., vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, vehicle driver interaction with a vehicle system, a rate of vehicle acceleration after stopping at a stoplight, etc.).

[0033] The apparatuses, systems and methods may determine transition of control for automated vehicles by conferring "driver signature" and image data. For example, the apparatuses, systems and methods may determine that a vehicle with automated piloting capabilities has transferred control from machine to human or from human to machine. The determination may be based on gesture recognition to identify a degree of driver interaction with steering wheel, or other control systems that enable transfer of control (e.g., hold down this lever, push this button, etc.). Vehicle dynamics may also be leveraged in order to confirm that a transfer has occurred.

[0034] A driver signature may be compiled by analyzing patterns of acceleration, braking, cornering, etc. For example, a driver signature may be based on a typical rate of acceleration after stopping at a stoplight. Each driver is likely to have a specific driving signature (i.e., each driver may produce a unique set of vehicle dynamics data). Fur-

thermore, an autonomous vehicle (i.e., self-driving systems) is likely to have a specific driving style which is different from human driver signatures. By conferring both gesture interaction with piloting controls (e.g., steering wheel) and driver signature, accurate determination of transfer of control may be achieved. The apparatuses, system and methods may communicate insurance rating decisions based on who is controlling the vehicle.

[0035] The apparatuses, systems and methods of the present disclosure may detect aggregations of driver distractions and may alter navigational directions of, for example, a navigation system to avoid distraction clusters. Driver movements within the vehicle (e.g., driver head pose, hand motions, body posture, etc.) may be detected. A category of driver behavior (e.g., two hands on the wheel, texting, talking on the phone, etc.) may be identified based on, for example, the driver movements within the vehicle. Vehicle dynamics may be analyzed, and situations in which the driver behavior is taking place may be determined. For example, a driver texting while stopped in a parking lot may be distinguished from the driver texting while making a left-hand turn. A current driver state for an individual vehicle (e.g., distracted, non-distracted, attentive, etc.) may be determined. A relative insurance risk score may be determined based on the driver state/behavior and the vehicle dynamics. The driver state and/or insurance risk score may be transmitted to a remote computing device. Notably, driver identity and profiles may be removed in order to provide aggregated and anonymous data. A vehicle location (at the time of distraction) may also be transmitted to the remote computing device.

[0036] A processor of a remote computing device may determine that there are multiple vehicles with risk scores that exceed a given threshold. Alternatively, or additionally, the processor may determine that there are multiple vehicles with driver states that are classified as potentially hazardous. The processor may determine that the vehicles with risk scores that exceed the threshold are within a given proximity of each other and/or at least one roadside equipment device. A group of distracted vehicles may be determined to be a cluster. The cluster may consist of a changing number of vehicles, depending on the risk score and proximities in real-time. For example, a vehicle in this distracted cluster is Vehicle A. The processor may analyze locations and headings of other vehicles, nearby the cluster of distracted vehicles. If the nearby vehicles are on a trajectory, or a route, as prescribed by an associated navigation system, and a vehicle that is approaching the cluster is Vehicle B, a processor onboard an approaching vehicle may reroute or redirect the driver of Vehicle B, including finding alternative routes, pathways, turns, etc. away from the distracted cluster. The driver of Vehicle B may elect to set this re-routing to automatic, selection-only, or deactivate completely.

[0037] Vulnerable road users may include bicyclists, pedestrians, and cars broken down on the side of the road. A processor may identify a location and/or trajectory of vulnerable road users. The processor may determine that the distracted cluster is approaching the vulnerable road users, or that the vulnerable road users are about to intersect trajectories with the distracted cluster. The processor may issue a warning to the vulnerable road user through a mobile device through at least one of an audio alert, a visual alert,

or a haptic alert. A warning may alert a vulnerable road user of a potential hazard in general, and/or of a specific hazard of the distracted cluster.

[0038] In a particular example, mobile phones of children may be coded differently such that if a child was about to jump into the street, the processor may generate a warning about a child pedestrian. Because known global positioning systems (GPS) include one-inch location precision, a standard GPS may enable enough precision for accurate prediction of the child's trajectory. The processor may alternatively, or additionally, warn drivers of bicyclists, pedestrians, and broken down vehicles on the side of the road. In another example, if a processor associated with individual A detects that the individual is distracted, and also detects that there is a school bus nearby, the processor may generate a warning to either the individual or a bus driver, or both.

[0039] Turning to FIG. 1, an example report 100, representative of vehicle in-cabin insurance risk evaluation, is depicted. The report 100 may include a title 105 (e.g., In-Cabin Risk Evaluation Report), a photograph of a driver 110, a name of a driver 111, and a drive identification 115 including, for example, a calendar date 116 and a time 117. The report 100 may also include value 121 (e.g., 67 centimeters) for a range of movement of most distinct postures 120. The report 100 is a chronological diagram 130 of various driver postures 129, 131, 132, 133, 134, 135 including details of a driver posture that the driver was in for the longest total time 125. The driver posture that the driver was in for the longest total time 125 may include a skeletal FIG. 126 representing the posture, a total elapsed time 127, and a number of individual occurrences of the posture 128. The report 100 may further include a graph 140 (e.g., a bar chart) including a title (e.g., posture vs. time distribution), a number of times a given posture was determined 142, and a total time in a given posture 143. The report 100 may also include the top five postures during an associated ride 145 including skeletal figures representative of the respective postures 150, 155, 160, 165, 170, a time in any given posture during an associated ride 151, 156, 161, 166, 171, and a number of occurrences of any given posture during an associated ride 152, 157, 162, 167, 172.

[0040] With reference to FIG. 2, a high-level block diagram of vehicle in-cabin system 200 is illustrated that may implement communications between a vehicle in-cabin device 205 and a remote computing device 210 (e.g., a remote server) to provide vehicle in-cabin device 205 location and/or orientation data, and vehicle interior occupant position data to, for example, an insurance related database 270. The vehicle in-cabin system 200 may acquire data from a vehicle in-cabin device 205 and generate three dimensional (3D) models of a vehicle interior and occupants within the vehicle interior. The vehicle in-cabin system 200 may also acquire data from a microphone 251, 252 and determine a source of sound and volume of sound within a vehicle interior.

[0041] For clarity, only one vehicle in-cabin device 205 is depicted in FIG. 2. While FIG. 2 depicts only one vehicle in-cabin device 205, it should be understood that any number of vehicle in-cabin devices 205 may be supported. The vehicle in-cabin device 205 may include a memory 220 and a processor 225 for storing and executing, respectively, a module 221. The module 221, stored in the memory 220 as a set of computer-readable instructions, may be related to a

vehicle interior and occupant position data collecting application that, when executed on the processor 225, causes vehicle in-cabin device location data to be stored in the memory 220. Execution of the module 221 may also cause the processor 225 to generate at least one 3D model of at least a portion of a vehicle occupant (e.g., a driver and/or passenger) within the vehicle interior. Execution of the module 221 may further cause the processor 225 to associate the vehicle in-cabin device location data with a time and, or date. Execution of the module 221 may further cause the processor 225 to communicate with the processor 255 of the remote computing device 210 via the network interface 230, the vehicle in-cabin device communications network connection 231 and the wireless communication network 215.

[0042] The vehicle in-cabin device 205 may also include a compass sensor 227, a global positioning system (GPS) sensor 229, and a battery 223. The vehicle in-cabin device 205 may further include an image sensor input 235 communicatively connected to, for example, a first image sensor 236 and a second image sensor 237. While two image sensors 236, 237 are depicted in FIG. 2, any number of image sensors may be included within a vehicle interior monitoring system and may be located within a vehicle interior. The vehicle in-cabin device 205 may also include an infrared sensor input 240 communicatively connected to a first infrared sensor 241 and a second infrared sensor 242. While two infrared sensors 241, 242 are depicted in FIG. 2, any number of infrared sensors may be included within a vehicle interior monitoring system and may be located within a vehicle interior. The vehicle in-cabin device 205 may further include an ultrasonic sensor input 245 communicatively connected to a first ultrasonic sensor 246 and a second ultrasonic sensor 247. While two ultrasonic sensors 246, 247 are depicted in FIG. 2, any number of ultrasonic sensors may be included within a vehicle interior monitoring system and may be located within a vehicle interior. The vehicle in-cabin device 205 may also include a microphone input 250 communicatively connected to a first microphone 251 and a second microphone 252. While two microphones 251, 252 are depicted in FIG. 2, any number of microphones may be included within a vehicle interior monitoring system and may be located within a vehicle interior. The vehicle in-cabin device 205 may further include a display/user input device 225.

[0043] As one example, a first image sensor 236 may be located in a driver-side A-pillar, a second image sensor 237 may be located in a passenger-side A-pillar, a first infrared sensor 241 may be located in a driver-side B-pillar, a second infrared sensor 242 may be located in a passenger-side B-pillar, first and second ultrasonic sensors 246, 247 may be located in a center portion of a vehicle dash and first and second microphones 251, 252 may be located on a bottom portion of a vehicle interior rearview mirror. The processor 225 may acquire position data from any one of, or all of, these sensors 236, 237, 241, 242, 246, 247, 251, 252 and generate at least one 3D model (e.g., a 3D model of at least a portion of a vehicle driver) based on the position data. The processor 225 may transmit data representative of at least one 3D model to the remote computing device 210. Alternatively, the processor 225 may transmit the position data to the remote computing device 210 and the processor 255 may generate at least one 3D model based on the position data. In either event, the processor 225 or the processor 255 retrieve data representative of a 3D model of a vehicle

operator and compare the data representative of the 3D model of at least a portion of the vehicle driver with data representative of at least a portion of the 3D model vehicle operator. The processor 225 and/or the processor 255 may generate a vehicle driver warning based on the comparison of the data representative of the 3D model of at least a portion of the vehicle driver with data representative of at least a portion of the 3D model vehicle operator to warn the vehicle operator that his position is indicative of inattentiveness. Alternatively, the processor 225 and/or the processor 255 may generate an advisory based on the comparison of the data representative of the 3D model of at least a portion of the vehicle driver with data representative of at least a portion of the 3D model of a vehicle operator to advise the vehicle operator how to correct her position to improve attentiveness.

[0044] The network interface 230 may be configured to facilitate communications between the vehicle in-cabin device 205 and the remote computing device 210 via any hardwired or wireless communication network 215, including for example a wireless LAN, MAN or WAN, WiFi, the Internet, or any combination thereof. Moreover, the vehicle in-cabin device 205 may be communicatively connected to the remote computing device 210 via any suitable communication system, such as via any publicly available or privately owned communication network, including those that use wireless communication structures, such as wireless communication networks, including for example, wireless LANs and WANs, satellite and cellular telephone communication systems, etc. The vehicle in-cabin device 205 may cause insurance risk related data to be stored in a remote computing device 210 memory 260 and/or a remote insurance related database 270.

[0045] The remote computing device 210 may include a memory 260 and a processor 255 for storing and executing, respectively, a module 261. The module 261, stored in the memory 260 as a set of computer-readable instructions, facilitates applications related to determining a vehicle in-cabin device location and/or collecting insurance risk related data. The module 261 may also facilitate communications between the computing device 210 and the vehicle in-cabin device 205 via a network interface 265, a remote computing device network connection 266 and the network 215 and other functions and instructions.

[0046] The computing device 210 may be communicatively coupled to an insurance related database 270. While the insurance related database 270 is shown in FIG. 2 as being communicatively coupled to the remote computing device 210, it should be understood that the insurance related database 270 may be located within separate remote servers (or any other suitable computing devices) communicatively coupled to the remote computing device 210. Optionally, portions of insurance related database 270 may be associated with memory modules that are separate from one another, such as a memory 220 of the vehicle in-cabin device 205.

[0047] Turning to FIG. 3A, a vehicle device 300a is depicted. The vehicle device 300a may be similar to, for example, the vehicle device 205 of FIG. 2. The vehicle device 300a may include a vehicle device registration module 310a, a reference image data receiving module 315a, an image sensor data receiving module 320a, a geographic information system (GIS) data receiving module 325a, a compass data receiving module 327a, a vehicle device

location/orientation module 329a, a day/time data receiving module 330a, a skeletal pose data generation module 335a, a vehicle telemetry system data receiving module 340a, a driver action prediction data generation module 345a, a driver action time stamp data generation module 350a, a driver action time stamp data transmission module 355a, a driver warning generation module 360a, and a report generation module 365a stored on a memory 305a as, for example, computer-readable instructions.

[0048] With reference to FIG. 3B, a vehicle device 300b is depicted. The vehicle device 300b may be similar to, for example, vehicle device 205 of FIG. 2. The vehicle device 300b may include a previously classified image data receiving module 315b, a current image data receiving module 320b, a vehicle dynamics data receiving module 325b, and a driver signature data generation module 330b stored on a memory 305b as, for example, computer-readable instructions.

[0049] Turning to FIG. 4 a remote computing device 400 is depicted. The remote computing device 400 may be similar to the remote computing device 210 of FIG. 2. The remote computing device 400 may include a reference image data transmission module 410, a driver action time stamp data receiving module 415, a driver action time stamp data analysis module 420, a report generation module 425, and a driver action time stamp data storage module 430 stored on a memory 405.

[0050] With reference to FIG. 5A, a flow diagram for an example method of registering a vehicle device (e.g., vehicle device 205, 300a, 300b) within a vehicle 500a is depicted. The method 500a may be implemented by a processor (e.g., processor 225) executing, for example, a portion of the modules 310a-365a of FIG. 3A. In particular, the processor 225 may execute a vehicle device registration module 310a and a reference image data receiving module 315a to cause the processor 225 to acquire image data from, for example, an image sensor (e.g., image sensor 265, 270 of FIG. 2) (block 505a). The processor 225 may further execute the vehicle device registration module 310a to cause the processor 225 to analyze the image sensor data to determine reference position of the vehicle device 205, 300a, 300b (block 510a). The processor 225 may further execute the vehicle device registration module 310a to cause the processor 225 to store data representative of the determined reference position of the vehicle driver (block 515a). The method 500a may be implemented, for example, in response to a driver of a vehicle placing a vehicle device 205, 300a, 300b within an associated vehicle (e.g., a driver may place the vehicle device 205, 300a, 300b on a dash of the vehicle near a passenger side A-pillar). Thereby, a generic vehicle module 205, 300a, 300b may be installed by a vehicle driver in any vehicle.

[0051] Vehicle driver postures may be rotated and scaled to be standardized (or normalized) vehicle device 205, 300a, 300b locations within a vehicle and standardized (or normalized) to an average human (i.e., applicable to all drivers). Subsequent to being registered within a given vehicle, a vehicle device 205, 300a, 300b may use image sensors 265, 270 to detect driver movements and record/categorize distinct driver postures (e.g., skeletal diagrams 125, 150, 155, 160, 165, 170). The methods and systems of the present disclosure may present results in two ways: 1) via detailed

report of different postures; and 2) via graphical representation of the postures detected with timeframe (e.g., as in report 100 of FIG. 1).

[0052] With reference to FIG. 5B, a flow diagram for an example method of generating data representative of a vehicle driver signature 500b is depicted. The method 500b may be implemented by a processor (e.g., processor 225) executing, for example, a portion of the modules 315b-340b of FIG. 3B. In particular, the processor 225 may execute the previously classified image data receiving module 315b to cause the processor 225 to, for example, receive previously classified image data (block 505b). The previously classified image data may be, for example, representative of images and/or extracted image features that have been previously classified as being indicative of a vehicle driver signature. More particularly, the previously classified image data may include images and/or extracted image features that have previously been classified as being representative of a vehicle occupant using a cellular telephone, a vehicle occupant looking out a vehicle side window, a vehicle occupant adjusting a vehicle radio, a vehicle occupant adjusting a vehicle heating, ventilation and air conditioning system, two vehicle occupants talking with one-another, a vehicle occupant reading a book or magazine, a vehicle occupant putting on makeup, a vehicle occupant looking at themselves in a mirror, etc. Alternatively, or additionally, the previously classified image data may, for example, be representative of known vehicle occupant locations/orientations, known cellular telephone locations/orientations, known vehicle occupant eye locations/orientations, known vehicle occupant head location/orientation, known vehicle occupant hand location/orientation, a known vehicle occupant torso location/orientation, a known seat belt location, a known vehicle seat location/orientation, etc.

[0053] The processor 225 may execute the current image data receiving module 320b to cause the processor 225 to, for example, receive current image data (block 510b). For example, the processor 225 may receive current image data from at least one vehicle sensor (e.g., at least one of a compass sensor 327, a GPS sensor 329, an image sensor 336, 337, an infrared sensor 341, 342, an ultrasonic sensor 346, 347, and/or a microphone 351, 352). The current image data may include images and/or extracted image features that are representative of a vehicle occupant (e.g., a vehicle driver) using a cellular telephone, a vehicle occupant looking out a vehicle side window, a vehicle occupant adjusting a vehicle radio, a vehicle occupant adjusting a vehicle heating, ventilation and air conditioning system, two vehicle occupants talking with one-another, a vehicle occupant reading a book or magazine, a vehicle occupant putting on makeup, a vehicle occupant looking at themselves in a mirror, etc. Alternatively, or additionally, the current image data may, for example, be representative of vehicle occupant locations/orientations, cellular telephone locations/orientations, vehicle occupant eye locations/orientations, vehicle occupant head location/orientation, vehicle occupant hand location/orientation, a vehicle occupant torso location/orientation, a seat belt location, a vehicle seat location/orientation, etc.

[0054] The processor 225 may execute the vehicle dynamics data receiving module 325b to cause the processor 225 to, for example, receive vehicle dynamics data from, for example, a vehicle system (block 515b). The vehicle dynamics data may be representative of a particular driving style,

or driving behavior, associated with a particular individual (e.g., vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, a vehicle driver interaction with a vehicle system, a vehicle driver interacting with a vehicle radio, a vehicle driver interacting with a cellular telephone, a vehicle driver interacting with a vehicle heating/air conditioning/ventilation system, a rate of vehicle acceleration after stopping at a stoplight, etc.).

[0055] The processor 225 may execute the driver signature data generation module 330b to cause the processor 225 to, for example, generate data representative of a vehicle driver signature (block 520b). For example, the processor 225 may compare the current image data with the previously classified image data and may determine that a current image and/or extracted image feature is representative of one of the previously classified images and/or extracted image features. Processor 225 may determine a vehicle driver signature based on a pattern of vehicle dynamics and driving style. Alternatively, or additionally, processor 225 may determine a driver signature based on vehicle interior image data and vehicle dynamics data. The vehicle interior image data may be representative of an identification of a particular individual. The vehicle dynamics data may be representative of a particular driving style, or driving behavior, associated with the particular individual.

[0056] The processor 225 may confirm an identity of a current driver of a given vehicle by conferring between at least two data streams (e.g., previously classified image data, current image data, and/or vehicle dynamics data). The image data may be, for example, either 3D point cloud data or 2D image or video data. The processor 225 may determine a “driver signature” based on a pattern of vehicle dynamics and driving style. Processor 225 may determine a vehicle driver signature based on a pattern of vehicle dynamics and driving style. Alternatively, or additionally, processor 225 may determine a driver signature based on vehicle interior image data and vehicle dynamics data. The vehicle interior image data may be representative of an identification of a particular individual. The vehicle dynamics data may be representative of a particular driving style, or driving behavior, associated with the particular individual. The vehicle dynamics data may be representative of a particular driving style, or driving behavior, associated with a particular individual (e.g., vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, a vehicle driver interaction with a vehicle system, a vehicle driver interacting with a vehicle radio, a vehicle driver interacting with a cellular telephone, a vehicle driver interacting with a vehicle heating/air conditioning/ventilation system, a rate of vehicle acceleration after stopping at a stoplight, etc.).

[0057] The processor 225 may determine transition of control for automated vehicles by conferring “driver signature” and image data. For example, the processor 225 may determine that a vehicle with automated piloting capabilities has transferred control from machine to human or from human to machine. The determination may be based on gesture recognition to identify a degree of driver interaction with a steering wheel, other control systems that enable transfer of control (e.g., hold down this lever, push this

button, etc.), that a vehicle occupant accepting control, or that a vehicle occupant transferring control. The processor 225 may also base a driver signature on vehicle dynamics in order to confirm that a transfer has occurred.

[0058] Processor 225 may determine a vehicle driver signature based on a pattern of vehicle dynamics and driving style. Alternatively, or additionally, processor 225 may determine a driver signature based on vehicle interior image data and vehicle dynamics data. The vehicle interior image data may be representative of an identification of a particular individual. The vehicle dynamics data may be representative of a particular driving style, or driving behavior, associated with the particular individual. The processor 225 may compile a driver signature by analyzing patterns of acceleration, braking, cornering, etc. For example, a driver signature may be based on a typical rate of acceleration after stopping at a stoplight. Each driver is likely to have a specific driving signature. An autonomous vehicle (i.e., a self-driving system) is likely to have a specific driving style which is different from human driver signatures. By conferring both gesture interaction with piloting controls (e.g., steering wheel) and driver signature, the processor 225 may achieve an accurate determination of transfer of control. The processor 225 may communicate insurance rating decisions based on who is controlling the vehicle as indicated by a determined driver signature.

[0059] The processor 225 may identify a given driver as an individual. For example, the processor 225 may identify the driver based on a known profile (e.g., a known profile that was previously downloaded to a memory within the system or a known profile calibrated previously). The processor 225 may determine a vehicle driver identification by identifying the driver from a known database (i.e., actually identifying the driver's name, or some other means such as a policy number, etc.).

[0060] Alternatively, or additionally, the processor 225 may determine a vehicle driver identification more relative. For example, assigned drivers of a vehicle may be stored within a system for a given insurance policy, and a known sample (e.g., a previously classified driver signature) of vehicle dynamics (i.e., a driving signature) and a corresponding digital image may be stored within a system for each driver associated with the insurance policy. The processor 225 may use the known "driver signature" to differentiate between different drivers assigned on that vehicle. For example, the processor 225 may identify a current driver by name. Alternatively, or additionally, the processor 225 may distinguish the current driver from the other drivers assigned to the insurance policy.

[0061] As a specific example, there may be three people assigned to a given vehicle on an associated insurance policy: Danny [person A]; Mary-Kate (twin) [person B]; and Ashley (twin) [person C]. While the processor 225 may not know their names, the processor may know person A, B and C. The processor 225 may establish Mary-Kate's unique driver signature (e.g., her digital image and her driving style) and may determine that the driver signature is person [B] as opposed to [A] or [C]. The next time that Mary-Kate drives, the processor 225 may attribute her driving history (e.g., behaviors and/or vehicle dynamics) to her [person B] as opposed to person C. While a given family may assign who is who manually, the processor 225 may keep every driver straight automatically (e.g., the processor 225 may automatically generate a unique driver signature) even if two

drivers are identical twins (i.e., a digital image may be the same or similar, however, driver dynamics may be different).

[0062] Turning to FIG. 6, a flow diagram of a method of generating data representative of a driver's action along with data representative of a time stamp 600 is depicted. The method 600 may be implemented by a processor (e.g., processor 225 of FIG. 2) executing, for example, at least a portion of the modules 310-365 of FIG. 3. In particular, the processor 225 may execute an image sensor data receiving module 320 to cause the processor 225 to receive image sensor data from an image sensor (e.g., image sensor 265, 270 of FIG. 2) (block 605). The processor 225 may further execute the image sensor data receiving module 320 to cause the processor 225 to receive point cloud data from an image sensor (e.g., image sensor 265, 270 of FIG. 2) (block 610). The processor 225 may execute a skeletal pose data generation module 335 to cause the processor 225 to process the point cloud data through, for example, a pose estimator to generate skeletal diagram data (block 615). The processor 225 may execute a reference image data receiving module 315 to cause the processor 225 to receive data representative of trained prediction modules (block 620). The processor 225 may execute a driver action prediction data generation module 345 to cause the processor 225 to compare the skeletal diagram data with the trained prediction models (block 620). The processor 225 may execute a day/time data receiving module 330 to cause the processor 225 to receive data representative of a day and/or time associated with a particular drive day/time (block 625). The processor 225 may execute a driver action time stamp data generation module 350 to cause the processor 225 to generate data representative of driver actions along with a timestamp of the action based on the driver action data generated in block 620 and further based on the data representative of the day/time (block 625) (block 630). The processor 225 may execute a driver action time stamp data transmission module 360 to cause the processor 225 to transfer the driver action time stamp data to, for example, a driver's cellular telephone via, for example, a Bluetooth communication (e.g., wireless transceiver 275 of FIG. 2). The method 600 may use skeleton tracking and face tracking technologies to identify different driver postures. Driver joints data points (e.g., joints data points 1806-1813 of FIG. 18) may be clustered to create entries which represent a unique driver posture. These postures may then be used for making predictions about the subject's driving habits. It should be understood that, in an event there are a plurality of distractions in, for example a given location, the processor 225 may aggregate a plurality of individual distractions to produce an aggregate distraction.

[0063] With reference to FIG. 7, and for prototype purposes, the system may implement a method to make predictions for a single driver 700. The method 700 may be implemented by a processor (e.g., processor 225 of FIG. 2) executing, for example, a portion of the modules 310-365 of FIG. 3. In particular, the processor 225 may execute an image sensor data receiving module 320 to cause the processor 225 to collect image data (block 705). The processor 225 may execute a skeletal pose data generation module 335 to cause the processor 225 to generate cluster data (block 710). The processor 225 may execute a driver action prediction data generation module 345 to predict driver's actions (block 715).

[0064] Turning to FIG. 8, a flow diagram for an example method of registering (or training) a vehicle device (e.g., vehicle device 205, 300) in a vehicle 800. The method may be implemented by a processor (e.g., processor 225 of FIG. 2) executing, for example, at least a portion of the modules 310-365 of FIG. 3. The method 800 may include receiving data points for a driver's skeletal diagram (block 805), initiating sensors and related programs (block 810), setting a sensor range to "near mode" (block 815), setting positioning to a "seated mode" (block 820), and instructing a driver on proper position for calibration (block 825) (e.g., driver should lean forward or move their hands/body (block 826)). The method 800 may also include polling the sensors (e.g., image sensors 265, 270) for driver initial position (block 830) and obtaining ten tracked points (e.g., points 1806-1813 of FIG. 18) (block 835). The method may further include instructing a driver to move to a normal seated position (block 840) and storing vehicle device registration data (block 845).

[0065] With reference to FIG. 9, a flow diagram for a method of categorizing various points associated with a driver's joints 900 (e.g., points 1806-1813 of FIG. 18, a driver's neck, a driver's shoulder, a driver's elbow, a driver's wrist, etc.) is depicted. The method 900 may include registering initial data points of a driver's skeleton diagram (block 905), saving all ten triplets associated with a driver's skeleton diagram and associated timestamp (block 910), finding nearest points for each point (block 915) (e.g., select nearest two vertical and nearest two horizontal points (block 916)). The method 900 may also include categorizing the highest points as a driver's head (e.g., point 1807 of FIG. 18) (block 920), categorizing the lowest two points as the driver's hands (e.g., points 1811, 1813 of FIG. 18) (block 925), and storing the categorized points (block 930).

[0066] Turning to FIG. 10, a flow diagram for an example method of predicting driver actions 1000 is depicted. The method 1000 may include tracking changes in the skeleton data points which are different than initial data points and record the changes in a database (block 1005), calculating average depth of ten initial points (block 1010), calculating variability percentage (block 1015) (e.g., variability sensitivity may differ depending on point and algorithms (block 1016)), draw ranges for ten joint positions (block 1020), and determine if a trip ended (block 1025). If the trip is determined to have ended (block 1025), the method includes saving the last position and ending the method 1000 (block 1030). If the trip is determined to not have ended (block 1025), the method 1000 checks a driver's current position vs. last logged position (range) (block 1035), and determines whether the driver's current position is new (block 1040). If the driver's current position is determined to be new (block 1040), the method 1000 saves all ten triplets and timestamps the triplets (block 1045), and then returns to block 1020. If the driver's current position is determined to not be new (block 1040), the method 1000 returns to block 1035.

[0067] Any one of the methods 800, 900, 1000, or portions thereof, may be used to identify a new driver (e.g., an algorithm for recognizing the driver being a new driver). The system may use the detailed algorithm mentioned as described in FIGS. 8-10. Alternatively, or additionally, any one of the methods 800, 900, 1000, or portions thereof, may be used to track movement of driver's upper body (e.g., an algorithm for tracking the movement of the driver's upper body is detailed in FIGS. 8-10). As yet a further alternative,

or addition, any one of the methods 800, 900, 1000, or portions thereof, may be used to log driver's clearly distinct postures at different times (e.g., an algorithm may identify and log distinct postures from the movements tracked). The methods and systems of the present disclosure may be implemented using C++. Associated application programming interfaces (APIs) and software development kits (SDKs) may support these platforms. Source code for the system may be controlled with, for example, versioning software available from Tortoise SVN.

[0068] With reference to FIG. 11, a sequence diagram for generating a vehicle in-cabin insurance risk evaluation report 1100 is depicted. A report generator 1105 may record/log a trigger 1111 at instance 1110. A data stream reader 1115 may identify a driver 1120 and record/log a trigger 1121. A data manipulation 1125 may match/create an entry 1126 and return a driver ID 1127. The data stream reader 1115 may read image sensor data 1130 and record/log a trigger 1131. The data manipulation 1125 may store snapshot data 1135 and record/log a trigger 1136. Cluster data 1140 may match a snapshot with an already registered cluster 1145 and may update cluster details 1146 at instance 1150. The report generator 1105 may get data and create a report 1156 at instance 1155. The data manipulation 1125 may return report data 1161 at instance 1160, and the report generator 1105 may print the report 1165.

[0069] Turning to FIG. 12, a detailed entity relationship (E-R) diagram 1200 is depicted. As depicted in FIG. 12, a driver 1230 and a device 1240 may be connected to a has info block 1205. The driver 1230 may be connected to a name 1231, a driver ID 1232, position coordinates 1233 (e.g., a face), a time stamp 1235, and a device ID 1236. The Device may be connected to a device ID 1241, a model 1242, and a manufacturer 1243. The driver 1230 and a ride 1245 may be connected to a takes block 1210. The ride 1245 may be connected to a ride ID 1246, an end time 1247, a vehicle 1248 (e.g., a car), a risk 1249, and a start time 1250. The ride 1245 and snapshots 1255 may be connected to a contains block 1215. The snapshots 1255 may be connected to a snapshots ID 1256, a ride ID 1257, and a time stamp 1258. The snapshots 1255 and a posture 1260 may be connected to a label with block 1220. The posture 126 may be connected to a posture ID 1261 and a time stamp 1262. The snapshots 1255 and joints 1275 may be connected to a consists of block 1225. The joints 1275 may be connected to a x-value 1266, a y-value 1267, a z-value 1268, a snapshot ID 1269, and a joint ID 1270.

[0070] With reference to FIG. 13, a method for creating a read-only database account 1300 is depicted. A database layer 1300 may be developed in MySQL server. The method 1300 may start (block 1305). All the rows in the database may be labeled as belonging to distribution G_1 (block 1310). The database creation 1300 may restart from a first row (block 1315). A probability that the row (1) dataset falls under distribution G_1 is obtained (blocks 1320, 1321). A probability that the row (2) dataset falls under distribution G_1 is obtained (blocks 1325, 1326). A categorization process 1330 may include finding a maximum probability 1331. If a probability that the row (1) is found to be highest (block 1330), the row is labeled with distribution G_1 (block 1332). If a probability that the row (2) is found to be highest (block 1330), a new G_2 is created and the row is labeled with distribution G_2 (block 1333) and the updated G_2 and associated parameters are stored in the database as a cluster

(block 1334). The method 1300 proceeds to the next row in the database (block 1335). A probability that the row (1) dataset falls under distribution G_1 , G_2 , . . . G_n is obtained (blocks 1340, 1341). A probability that the row (2) dataset falls under a new distribution is obtained (blocks 1345, 1346). A categorization process 1350 may be similar to the categorization process 1330. A determination as to whether the current row is the end of the database is made (block 1355). If the current row is determined to not be the last row (block 1355), the method 1300 returns to block 1335. If the current row is determined to be the last row (block 1355), the method 1300 proceeds to determine if the process discovered a new G_s or updated existing ones (block 1360). If the process is determined to have discovered a new G_s or updated existing ones (block 1360), the method 1300 returns to block 1315. If the process is determined to not have discovered a new G_s or updated existing ones (block 1360), all the existing clusters may be identified and results may be printed (block 1365) and the method 1300 ends (block 1370).

[0071] Turning to FIG. 14, a high-level block diagram of a development environment 1400 is depicted. The development environment 1400 may include an image sensor 1410 and a server 1405 hosting a database 1415 and VC++ implementation for collecting and clustering data. A user interface of the development environment may have a model car, parked car, or a dummy setup for a user to act as a driver. The system may analyze the movements of the driver during a trial period and may generate two sets of reports: 1) A live video of the skeleton frames with start, end and total time for the ride demo; and 2) A report shown also as charts of different postures and time spent for each posture as depicted, for example, in FIG. 1. The development environment is focused on building a working model of the concept. The end-to-end system uses Microsoft Kinect, Microsoft Visual Studio C++, MySQL database and Microsoft Windows as platform.

[0072] With reference to FIG. 15, a system diagram 1500 is depicted for a development environment of FIG. 14. The system 1500 may include HTML and/or GUI APIs 1505, a MYSQL database 1510, and SigmaNI+Open NI SDKs 1515. The system diagram 1500 depicts different C++ modules for different functionalities of the project. The system 1500 may also include an AppComponents::iDataManipulation component 1525 to interact with the MYSQL database 1510. All other components may use APIs in this component to interact with MYSQL database. The system 1500 may further include an AppComponents::iReadStream component 1535 to interact with Sensor hardware via KinectSDK middleware (e.g., SigmaNI+Open NI SDKs 1515). The iReadStream component 1535 may read a data stream from the sensor (e.g., image sensor 260, 265 of FIG. 1) and may store the data structure in a Snapshot table for further clustering and processing. The system 1500 may also include an AppComponents::iClusterData component 1530 that may read snapshot data stored by the iReadStream component 1535 and may cluster the data to identify driver postures. The AppComponents::iClusterData component 1530 may begin to function once new data is stored in a database by the iReadStream component 1535. The system 1500 may further include an AppComponents::iPredictionModule component 1540 that may function as a prediction engine, and may have algorithms to implement driving habit analysis for the captured data. The system 1500

may also include an AppComponents::iReportGenerator component 1520 that, for successful demonstration, a report will be generated. The AppComponents::iReportGenerator component 1520 may have APIs to read the data via iDataManipulation component 1525 from the database and generate report. This component will also display the live video of the participant on the screen. For the live video, it will capture the data directly from iReadStream component 1535.

[0073] An AppComponents::iDataManipulation 1525 may include input related to business objects acquired from or required by various business methods in other components. Output/Service may be provided for business objects extracted from a database via data access objects and methods. Depending on which component is calling, this component may have generic and client specific APIs for serving various business objects. Component/Entity process: Data connection; Connection pool; DAOs for below entities; Driver; Snapshot Object; RideDetails; and PosturesDetails. Constraints may include initial connection pool size of ten and max size may be thirty.

[0074] An AppComponents::iReadStream component 1535 may include input for an event to start and stop reading a video and sensor data stream from hardware. A SDK APIs may be used for reading skeleton, face and hand tracking data. Output/Service may be provided via snapshot objects and relevant joints coordinates may be output and stored in the database using Data manipulation component 1525. Live data may be transported to ReportGenerator component 1520. Component/Entity process may work as a batch process to start and stop logging the read data in the database when triggered. The component also needs to be able to transmit live data to iReportGenerator component 1520 to show it on screen. Constraints may include appropriate buffering and error handling which may be done, to make sure appropriate error messages are displayed/captured for downstream components.

[0075] An AppComponents::iClusterData component 1530 may input snapshot data read from iReadStream and a database. Output/Service may be provided and assign a postureID to a snapshot and update the posture-database. Component/Entity process may include: Retrieving snapshot and posture information from database; Matching snapshots with postures; Inserting new snapshot/posture information to database; Implementations of unsupervised clustering algorithms. Constraints may include a number of clusters generated has a limit.

[0076] An AppComponents::iPredictionModule component 1540 may serve to take in data from a database, and turn the data into information to leverage. The AppComponents::iPredictionModule component 1540 may identify risky drivers, review their in-cabin driving habits, and eventually act to curb these risky habits. This section explains how the data may be modeled to better understand which factors correlate to a defined risk metric and how certain behavior patterns contribute to a higher insurance risk rating.

[0077] An AppComponents::iReportGenerator 1520 may include input information taken from a database, the ten coordinates taken from the data stream during a demo, a start time, an elapsed time and some dummy information. Output/Service may be provided including a video of skeleton frames with start time and elapsed time and a report that displays charts that may illustrate what happened during the demo. The report may include a picture of the driver, the

driver's name, and the range of movement of most distinct postures. The report may also have a line graph and a bar graph that show how much time the driver spent in each posture. The report may display the skeleton coordinates of the five postures the driver was in the most along with the time and number of occurrences of each. Component/Entity process may include: a Generator; a Report; a Video; a DAOs for below entities; a Ride; a Posture and a Joint. Constraints may include a demo that may have at least five different postures. Number of postures and number of occurrences should not exceed max array length.

[0078] Turning to FIG. 16, a system for generating data representative of a vehicle in-cabin insurance risk evaluation 1600 is depicted. The system 1600 may include a plurality of vehicle devices 1605 communicatively coupled to a data processing, filtering and load balancing server 1615 via a wireless webservice port 1610 to send and receive data. The system 1600 may also include a database server 1620 and database 1621 to store in-cabin data, and an algorithm server 1625 to continuously refine algorithm data and an associated prediction engine. When multiple sensors are used, a SigmaNI wrapper may be used as an abstraction layer for code. This may ensure that if a sensor is changed, or different sensors are user, minimal code changes are required.

[0079] With reference to FIG. 17, when SigmaNI is not an approved software, an implementation 1700 may directly interact with a SDK 1710 to get the driver data from a sensor 1705 for generation data representative of vehicle in-cabin insurance risk evaluations 1715 and storing the data in a database 1720. The system 1700 may use sensors (e.g., image sensor 260, 265 of FIG. 1) for detecting the driving postures, such as provided by Microsoft Kinect for windows, Carmine 1.09 and/or Softkinect DS325. The following SDKs may be used with the above hardware: a Kinect SDK, an OpenNI, a Softkinect SDK and/or a SigmaNI.

[0080] Turning to FIG. 18, a posture (or skeletal diagram) 1800 may include eight joint positions 1806-1813 for a driver's upper body 1805. An associated cluster may include ten rounds with radius 10 cm and centered at ten 3-dimensional points. A match (posture p, cluster c) may return true if all the ten joint positions of the posture are contained in the ten balls for the cluster accordingly, otherwise returns false. A distance of two points may be measured using a Euclidean distance. For example, given a pair of 3-D points, $p=(p_1, p_2, p_3)$ and $q=(q_1, q_2, q_3)$: $\text{distance}(p, q)=\sqrt{(p_1-q_1)^2+(p_2-q_2)^2+(p_3-q_3)^2}$. A cube in 3-dimentional space consists of all points (x, y, z) that satisfy the following conditions: $a \leq x \leq b$, $c \leq y \leq d$, $e \leq z \leq f$, where $b-a=d-c=f-e$. When initialization: a first cluster may be defined by the ten joint positions of the first posture. A cluster may be added to the initial cluster list, denote CL Loop: for each of subsequent postures, say P, for each cluster in CL, say C, if Match (P, C): Label P with C, break, End For. If P does not have a cluster label, create a new cluster C' and add C' to CL—End For and BR4 [TR 4.1, 4.2, 4.3 and 4.4] Create risk profile of the driver.

[0081] With reference to FIG. 19, an object design for a detailed entity relationship (E-R) diagram 1900 is depicted. An associated database layer may be developed in MySQL server. The entity relationship 1900 may include a device 1905 connected to a driver 1910, connected to a ride 1920, connected to a snapshot 1930 which is connected to both joints 1935 and a posture 1945. The device 1905 may include a device ID 1906, a model, 1907 and a manufacturer

1908. The driver 1910 may include a driver ID 1911, a device ID 1912, a name 1913, face coordinates 1914, and a time stamp 1915. The ride 1920 may include a ride ID 1921, a driver ID 1922, a start time 1923, an end time 1924, a car 1925, and a risk 1920. The snapshot may include a snapshot ID 1931, a ride ID 1932, and a time stamp 1933. The joints 1935 may include a joint ID 1936, a snapshot ID 1937, a x-value 1938, a y-value 1939, and a z-value 1940. The posture 1945 may include a posture ID 1946, a snapshot ID 1947, and a time stamp 1948.

[0082] Turning to FIG. 20, a class diagram 2000 may include a BaseDAO 2005, a DeviceDAO 2010, a DriverDAO 2015, a SnapshotDAO 2015, a JointDAO 2035, a PostureDAO 2045, and a RideDAO 2050. The BaseDAO 2005 may include a con: DBConnection 2006 and a #getConnection() 2007. The DeviceDAO 2010 may include a DeviceID: String 2011, a Model: String 2012, a Manufacturer: String 2013, and a getters/setters 2014. The DriverDAO 2015 may include a DriverID: String 2016, a Name: String 2017, a FaceCoordinates: Array(int 100) 2018, a Device Obj: Device DAO 2019, a timestamp: timestamp 2020, and a getters/setters 2012. The SnapshotDAO 2015 may include a SnapshotID: String 2026, a RideID: String 2027, a TimeStamp: timestamp 2028, a Joints: Array (jointDAO 10) 2029, and a getters/setters 2030. The JointDAO 2035 may include a JointID: String 2036, a X: int 2037, a Y: int 2038, a Z: int 2039, and a getters/setters 2040. The PostureDAO 2045 may include a PostureID: String 2046, a SnapshotID: String 2047, a getters/setters 2048, and a fetTopPostureIDs (Postures) 2049. The RideDAO 2050 may include a RideID: String 2051, a DriverObj: DriverDAO 2052, a StartTime: timestamp 2053, an EndTime: timestamp 2054, and a getters/setters 2055.

[0083] With reference to FIG. 21, a class diagram 2100 may include a ReadSensorData component 2105, a ReadKinectSensorData component 2115, and a ReadCarmineSensorData component 2120. The ReadSensorData component 2105 may include a Snapshot: SnapshotDAO 2106, an initialize() parameter 2107, a readDataStream() parameter 2108, a saveSnapshot() parameter 2109, a transmitLive() parameter 2110, and a getters/setter parameter 2111. The ReadKinectSensorData component 2115 may include an initializeKinect() parameter 2116. The ReadCarmineSensorData component 2120 may include an initializeCarmine() parameter 2121.

[0084] Turning to FIG. 22, a class diagram 2200 may include a ClusterData component 2205, a Posture component 2225, a Snapshot component 2230, and a Joint component 2235. The ClusterData component 2205 may include a surSS: SnapShot 2206, a postures: ArrayList(Posturess) 2207, a postureID integer 2208, a Match_Posture() 2209, a Update_DB() 2210, and a getters/setters 2211. The Posture component 2225 may include a postureID: integer 2226, a Cneters:Array(Joint) 2227, a Radius: Number 2228, and a getters/setters 2229. The Snapshot component 2230 may include a SnapshotID: Integer 2231, a Joints: Array(Joint) 2232, a timestamp: Number 2233, and a getters/setters 2234. The Joint component 2235 may include a x: Number 2236, a y: Number 2237, a z: Number 2238, and a getters/setters 2239.

[0085] With reference to FIG. 23, a class diagram 2300 may include a ReportGenerator component 2305, a Report component 2310, and a Video component 2320. The ReportGenerator component 2305 may include a Report: Report

2306, a Video: Video **2307**, and a getters/setters **2308**. The Report component **2310** may include a DriverName: String **2311**, a RideObj: RideDAO **2312**, a getters/setters **2313**, a drawLineGraph(RideObj) **2314**, a drawBarGraph(RideObj) **2315**, and a drawTopPostures(RideObj) **2316**. The Video component **2320** may include a CurrentPosture: PostureDAO **2321**, a StartTime: timestamp **2322**, a CurrentTime: timestamp **2323**, a getters/setters **2324**, a displayPosture(CurrentPosture) **2325**, and a displayTimes(startTime, currentTime) **2326**.

[0086] A car-sharing insurance product could more specifically insure the driver, regardless of the car. Traditional underwriting looks at the driver-vehicle combination. What car-sharing would allow you to do is to more heavily weight the risk of the driver alone. The methods and systems of the present disclosure may allow car-sharing to get that risk information on the driver and carry it forward to whatever car they use. This would be tailored for that particular driver's behavior, rather than demographic and vehicle-use factors. This would allow certain car-sharing entities to have a cost advantage. If they are paying less insurance—or more specific insurance—they could pass those savings to their customers and have a retention strategy.

[0087] The methods and systems of the present disclosure may allow emergency responders to, for example, use gesture recognition systems from an aftermarket/insurance device in order to provide an estimate to first responders about the severity of the crash and what kinds of resources/equipment/expertise is required in order to extricate. Using the gesture recognition systems from an aftermarket/insurance device in order to provide an estimate to first responders about the severity of the crash and what kinds of resources/equipment/expertise is required in order to triage—have some idea of what emergency medical needs could be upon arrival. Since the “golden hour” is so critical, and it's not always known how much of that hour has already expired, even a preliminary or broad clue could be helpful in the triage process. The aftermarket gesture recognition device is already operating at the time of the crash. It is collecting data about the driver's position/posture and the location of the arms relative to the body and structures in the vehicle (i.e. the steering wheel). Accelerometers in the device are able to recognize that a crash has occurred (if a pre-determined acceleration threshold has been reached). Upon crash detection the device could transmit via the driver's phone (which is already connected via Bluetooth) or perhaps transmit using an onboard transmitter that uses emergency frequencies (and therefore does not require consumer to pay for data fees). Using gesture recognition from any original equipment or aftermarket gesture tracking device, whether or not for insurance purposes.

[0088] The methods and systems of the present disclosure may allow for Transition from Automated to Manual Driving Mode in the case of vehicle automation systems operating the piloting functions with the human in a supervisory role. The vehicle encounters a situation where it needs to transfer control to the driver, but the driver may or may not be ready to resume control. The methods and systems of the present disclosure may allow gesture recognition systems, or any gesture recognition system, to be used to determine if the driver is ready to resume control. If he/she is not ready, then get his/her attention quickly. The gesture recognition would be used to ascertain whether the driver is ready to resume control by evaluating the driver's posture, the loca-

tion of hands, the orientation of head, body language. Use machine learning to evaluate driver engagement/attention/readiness-to-engage based on those variables. The gesture recognition could be any original in-vehicle equipment or aftermarket device.

[0089] The methods and systems of the present disclosure may distinguish between Automated and Manual driving modalities for variable insurance rating for a scenario where there are many vehicles that are capable of automatically operating the piloting functions, and are capable of the driver manually operating the piloting functions. The driver can elect to switch between automated and manual driving modes at any point during a drive. Gesture recognition would be utilized to distinguish whether a driver is operating the vehicle manually, or whether the vehicle is operating automatically. This could be determined through either OEM or aftermarket hardware. The sensors and software algorithms are able to differentiate between automatic and manual driving based on hand movements, head movements, body posture, eye movements. It can distinguish between the driver making hand contact with the steering wheel (to show that he/she is supervising) while acting as a supervisor, versus the driver providing steering input for piloting purposes. Depending on who/what is operating the vehicle would determine what real-time insurance rates the customer is charged.

[0090] The methods and systems of the present disclosure may provide a tool for measuring driver distraction where gesture recognition may be used to identify, distinguish and quantify driver distracted for safety evaluation of vehicle automation systems. This would be used to define metrics and evaluate safety risk for the vehicle human-machine interface as a whole, or individual systems in the case where vehicles have automation and vehicle-to-vehicle/vehicle-to-infrastructure communication capabilities. Where Vehicle automation: the vehicle is capable of performing piloting functions without driver input. Where Vehicle-to-vehicle/vehicle-to-infrastructure communication: the vehicle is capable of communicating data about the first vehicle dynamics or environmental traffic/weather conditions around the first vehicle. For any entity looking to evaluate the safety or risk presented by a vehicle with automated driving capabilities, gesture recognition could be useful to quantify risk presented by driver distraction resulting from any vehicle system in the cabin (i.e. an entertainment system, a feature that automates one or more functions of piloting, a convenience system). With the rise of vehicle automation systems and capabilities, tools will be needed to evaluate the safety of individual systems in the car, or the car as a whole. Much uncertainty remains about how these systems will be used by drivers (especially those who are not from the community of automotive engineering or automotive safety). Determining whether they create a net benefit to drivers is a big question. The methods and systems of the present disclosure may allow gesture recognition could be used to identify the presence of distracted driving behaviors that are correlated with the presence of vehicle automation capabilities. The distracted could be quantified by duration that the driver engages in certain behaviors. Risk quantification may also be measured by weighting certain behaviors with higher severity than other behaviors, so the duration times are weighted. Risk quantification may also differentiate subcategories of behaviors based on degree of motion of hands, head, eyes, body. For example, the methods and

systems of the present disclosure may distinguish texting with the phone on the steering wheel from texting with the phone in the driver's lap requiring frequent glances up and down. The latter would be quantified with greater risk in terms of severity of distraction. The purpose of this risk evaluation could be for reasons including but not limited to adhere to vehicle regulations, providing information to the general public, vehicle design testing or insurance purposes. [0091] This detailed description is to be construed as exemplary only and does not describe every possible embodiment, as describing every possible embodiment would be impractical, if not impossible. One may be implement numerous alternate embodiments, using either current technology or technology developed after the filing date of this application.

What is claimed is:

1. A device for dynamically rerouting vehicle navigation, the device comprising:

a processor; and

a memory including instructions that, when executed by the processor, cause the processor to:

receive an indication from a first vehicle, the indication including a location of the first vehicle and a risk score of the first vehicle, wherein the risk score is calculated based on vehicle dynamics data of the first vehicle and a driver signature of a driver of the first vehicle, the driver signature identifying a driver behavior;

detect the risk score of the first vehicle exceeds a predetermined threshold; and

reroute navigation of a second vehicle to avoid the first vehicle.

2. The device of claim 1, wherein the instructions, when executed by the processor, further cause the processor to:

detect a cluster of one or more vehicles, the cluster of one or more vehicles having risk scores higher than the predetermined threshold, the cluster of one or more vehicles including the first vehicle, and the cluster of one or more vehicles within a geographic proximity of each other; and

reroute navigation of the second vehicle to avoid the cluster of one or more vehicles.

3. The device of claim 2, wherein the instructions, when executed by the processor, further cause the processor to:

receive, in real-time, updated location data of the cluster of one or more vehicles; and

update, a location of the cluster of one or more vehicles based on the received location data.

4. The device of claim 2, wherein the instructions, when executed by the processor, further cause the processor to:

receive, in real-time, updated risk scores of the one or more vehicles; and

update, a number of vehicles in the cluster of one or more vehicles based on the received risk scores.

5. The device of claim 1, wherein the driver signature is generated based on vehicle dynamics data and a comparison of vehicle interior data captured by one or more sensors with previously classified vehicle interior data.

6. The device of claim 5, wherein the vehicle dynamics data is representative of at least one of: vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, vehicle

driver interaction with a vehicle system, or a rate of vehicle acceleration after stopping at a stoplight.

7. The device of claim 1, wherein the instructions, when executed by the processor, further cause the processor to:

receive a user input from a driver of the second vehicle to select a mode for navigation rerouting, wherein the mode corresponds to at least one of: automatically rerouting, requesting confirmation from the driver of the second vehicle prior to rerouting, and disabling rerouting.

8. A computer-implemented method for dynamically rerouting vehicle navigation, the computer-implemented method comprising:

receiving, at a processor, an indication from a first vehicle, the indication including a location of the first vehicle and a risk score of the first vehicle, wherein the risk score is calculated based on vehicle dynamics data and a driver signature of a driver of the first vehicle, the driver signature identifying a driver behavior;

detecting, by the processor, the risk score of the first vehicle exceeds a predetermined threshold; and

rerouting, by the processor, navigation of a second vehicle to avoid the first vehicle.

9. The computer-implemented method of claim 8, further comprising:

detecting, by the processor, a cluster of one or more vehicles, the cluster of one or more vehicles having risk scores higher than the predetermined threshold, the cluster of one or more vehicles including the first vehicle, and the cluster of one or more vehicles within a geographic proximity of each other; and

reroute navigation of the second vehicle to avoid the cluster of one or more vehicles.

10. The computer-implemented method of claim 9, detecting the cluster of one or more vehicles further comprises:

receiving, by the processor and in real-time, updated location data of the cluster of one or more vehicles; and

updating, by the processor, a location of the cluster of one or more vehicles based on the received location data.

11. The computer-implemented method of claim 9, wherein detecting the cluster of one or more vehicles further comprises:

receiving, by the processor and in real-time, updated risk scores of the one or more vehicles; and

updating, by the processor, a number of vehicles in the cluster of one or more vehicles based on the received risk scores.

12. The computer-implemented method of claim 8 wherein the driver signature is generated based on vehicle dynamics data and a comparison of vehicle interior data captured by one or more sensors with previously classified vehicle interior data.

13. The computer-implemented method of claim 12, wherein the vehicle dynamics data is representative of at least one of: vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, vehicle driver interaction with a vehicle system, or a rate of vehicle acceleration after stopping at a stoplight.

14. The computer-implemented method of claim 8, further comprising:

receiving, by the processor, a user input from a driver of the second vehicle to select a mode for navigation rerouting, wherein the mode corresponds to at least one of: automatically rerouting, requesting confirmation from the driver of the second vehicle prior to rerouting, and disabling rerouting.

15. A non-transitory computer-readable medium storing computer-readable instructions that, when executed by a processor, cause the processor to:

receive an indication from a first vehicle, the indication including a location of the first vehicle and a risk score of the first vehicle, wherein the risk score is calculated based on vehicle dynamics data of the first vehicle and a driver signature of a driver of the first vehicle, the driver signature identifying a driver behavior;

detect the risk score of the first vehicle exceeds a predetermined threshold; and

reroute navigation of a second vehicle to avoid the first vehicle.

16. The non-transitory computer-readable medium of claim 15, wherein the computer-readable instructions further cause the processor to:

detect a cluster of one or more vehicles, the cluster of one or more vehicles having risk scores higher than the predetermined threshold, the cluster of one or more vehicles including the first vehicle, and the cluster of one or more vehicles within a geographic proximity of each other; and

reroute navigation of the second vehicle to avoid the cluster of one or more vehicles.

17. The non-transitory computer-readable medium of claim 16, wherein the computer-readable instructions further cause the processor to:

receive, in real-time, updated location data of the cluster of one or more vehicles; and
update, a location of the cluster of one or more vehicles based on the received location data.

18. The non-transitory computer-readable medium of claim 16, wherein the computer-readable instructions further cause the processor to:

receive, in real-time, updated risk scores of the one or more vehicles; and

update, a number of vehicles in the cluster of one or more vehicles based on the received risk scores.

19. The non-transitory computer-readable medium of claim 15, wherein:

the driver signature is generated based on vehicle dynamics data and a comparison of vehicle interior data captured by one or more sensors with previously classified vehicle interior data; and

the vehicle dynamics data is representative of at least one of: vehicle acceleration, patterns of vehicle acceleration, vehicle braking, patterns of vehicle braking, vehicle steering, patterns of vehicle steering, vehicle cornering, patterns of vehicle cornering, vehicle driver interaction with a vehicle system, or a rate of vehicle acceleration after stopping at a stoplight.

20. The non-transitory computer-readable medium of claim 15, wherein the computer-readable instructions further cause the processor to:

receive a user input from a driver of the second vehicle to select a mode for navigation rerouting, wherein the mode corresponds to at least one of: automatically rerouting, requesting confirmation from the driver of the second vehicle prior to rerouting, and disabling rerouting.

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